

Commented [PA1]: Comment (29) on odd page footer by PAUL Alexander (19 Dez 2025 11:18)
2025 soon to be updated to 2026

SYSTEM REQUIREMENTS SPECIFICATION DIL

Digital Interlocking OBB
Data Center & Networks

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Document Identifier Radical	DTC	Enterprise Identifier	Revision
N/A	305	16282~0241	01 PD01

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APPROVAL		
	Name and Role/Job Title	E-Signature
Written by	Eber G. / RM	
Verified by	Rotter F. / QAM	
Approved by	Eichberger M. / PDA	

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LOG OF CHANGES			
Edition	Date	Author	Change Description
01 PD01	2025-12-xx	Hitachi Rail GTS Austria / EJT RM G. Eber	First edition. Details see 4.1.1

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1. INTRODUCTION

1.1 PURPOSE

The present document contains the system requirements of the Digital Interlocking (DIL). The system is developed in the context of the "Generic Application Digital Interlocking OBB". This part of the SSS focuses on the **Data Center** requirements derived from [PH_PART_D4] and the **Network** requirements derived from [PH_PART_D6].

1.2 INTENDED AUDIENCE

- Product Design Authority
- Technical Project Managers
- Solution Engineering Managers
- Product Line Manager
- Test Manager
- Validator
- Verifier
- Safety Assurance Manager
- Security Assurance Manager
- Quality Assurance Manager
- RAM Assurance Manager
- System Architects

1.3 DISTRIBUTION

A new version of the document is distributed to all persons according to chapter 1.2

1.4 DOCUMENT MANAGEMENT

See Requirements Management Plan [DIL_RMP] for a description of the SSS document management.

1.5 REFERENCED DOCUMENTS

[SP_GLOSS]	SharePoint Project Glossary Abbreviations for Glossary without duplicates.xlsx
[EL_GLOSS]	Glossary: Definitions, Acronyms and Abbreviations - ELEKTRA 2.x System, 3BU 15000 0081 DCAPA, Ed. 06
[DIL_RMP]	Requirements Management Plan – Digital Interlocking OBB, 3BU 17300 1207 DUAPA, Ed. 01

Commented [BD2]: Comment (17) by BAUMGARTNER Dieter (18 Dez 2025 8:39)
Solche Links sind nicht nachhaltig

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[LH_PART_D]	DSTW Streckenausrüstung OBB - Teil D Leistungsbeschreibung und Lastenhefte, 3BU 17300 0004 DSAPC, Ed. 0405
[PH_PART_D2.2]	Pflichtenheft Digital Interlocking OBB - D2.2 Teilpflichtenheft RAMS, 3BU 17300 0022 DTAPC, Ed. 02
[PH_PART_D2.3]	Pflichtenheft Digital Interlocking OBB - D2.3 Teilpflichtenheft Allgemeine Anforderungen, 3BU 17300 0023 DTAPC, Ed. 02
[PH_PART_D2.5]	Pflichtenheft Digital Interlocking OBB - D2.5 Teilpflichtenheft Teilsystem DSTW-ZE, 3BU 17300 0025 DTAPC, Ed. 02
[PH_PART_D2.6]	Pflichtenheft Digital Interlocking OBB - D2.6 Teilpflichtenheft Teilsystem Weiche, 3BU 17300 0026 DTAPC, Ed. 02
[PH_PART_D4]	Pflichtenheft Digital Interlocking OBB – D4 Teilpflichtenheft Rechenzentrum, 3BU 17300 0034 DTAPC, Ed. 01
[PH_PART_D6]	Pflichtenheft Digital Interlocking OBB - D6 Teilpflichtenheft Netze, 3BU 17300 0036 DTAPC, Ed. 01
[CMP]	Configuration Management Plan, 3BU 15000 3338 QMAPA, Ed.11
[TECHNICAL_CONCEPTS]	Digitales Stellwerk - Technische Konzepte, 3BU 17300 0010 DSAPC, Ed.04
[ARCHITECTURE_DRAWING]	Top Level Architecture Drawing - Generic Application - Digital Interlocking OBB, 3BU 17300 1218 DSAPA, Ed. 0102
[BFZ_AUSFALLKONZEPT]	Anforderungsspezifikation BFZ-Ausfallszenarien Leit- und Sicherungstechnik, 3BU 51500 2003 DSAPC, Ed.01
[RAM]	Project specific RAM Plan - Generic Application Digital Interlocking OBB - Solution Element VIL 3BU_17300_1209_DUAPA, Ed.01
[R10]	Eisenbahnanlagen Schwerpunktkonzept aus Sicht des Arbeitnehmer/innenschutzes
[RW 13.01.01]	Regelwerk Eisenbahnsicherungsanlagen
[RW 13.01.02]	Regelwerk Punktförmige Zubebeeinflussung
[RW 30.02]	Regelwerk Signalbuch
[RW 47.07.10]	Regelwerk Eisenbahnanlagen
[SSS_DIL_GENERAL]	System Requirements Specification DIL – General Requirements, 3BU 17300 1251 DTAPA Ed.01
[SSS_DIL_DATA_CENTER]	System Requirements Specification DIL - Data Center, 3BU 17300 126x DTAPA Ed.01 (Planned)
[SSS_DIL_SECURITY]	System Requirements Specification DIL - Security, 3BU 17300 1252 DTAPA Ed.01 (Planned)(in review)
[SSS_DIL_CU_1]	System Requirements Specification DIL – Central Unit – Part 1, 3BU 17300 1255 DTAPA Ed.01
[SSS_DIL_CU_2]	System Requirements Specification DIL – Central Unit – Part 2, 3BU 17300 1258 DTAPA Ed.01 (Planned)
[SSS_DIL_CU_3]	System Requirements Specification DIL – Central Unit – Part 3, 3BU 17300 1259 DTAPA Ed.01 (Planned)
[Low_Voltage]	Low Voltage Directive 2014/35/EU
[EMC]	EMC Directive 2014/30/EU
[RoHS 2]	Directive on hazardous substances 2011/65/EU (RoHS 2)
[Ecodesign]	Ecodesign Directive 2019/424/EU (Lot 9)

Commented [SE3]: Proposed Change (42) by STEMMER Erich (30 Dez 2025 15:08)
neueste Version

Commented [SE4]: Comment (43) by STEMMER Erich (30 Dez 2025 15:10)
Edition 5 sollte mit Mitte Jänner verfügbar sein

Commented [SE5]: Proposed Change (46) by STEMMER Erich (30 Dez 2025 15:18)
aktuelle Version ED02

Commented [SE6]: Comment (45) by STEMMER Erich (30 Dez 2025 15:15)
Ich dachte das ist dieses Dokument !?

Commented [SE7]: Proposed Change (44) by STEMMER Erich (30 Dez 2025 15:13)
let's check for a version in DWF

Commented [SE8]: Comment (47) by STEMMER Erich (30 Dez 2025 15:22)
Diese Nummer ist nun für Datacenter vergeben -> siehe dieses Dokument

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[EN 1991-1-4]	1991-1-4:2023 Eurocode 1: Einwirkungen auf Tragwerke Teil 1-4: Allgemeine Einwirkungen - Windlasten Nationale Festlegungen zu ÖNORM EN 1991-1-4 und nationale Ergänzungen 2023-04-01
[EN 50121-1]	EN 50121-1:2017 Bahnanwendungen - Elektromagnetische Verträglichkeit -- Teil 1: Allgemeines 2017-12
[EN 50121-4]	EN 50121-4:2016 +A1:2019 Railway applications – Electromagnetic compatibility - Part 4: Emission and immunity of the signalling and telecommunications apparatus
[EN 50122]	EN 50122-1:2011 +A1:2011 +AC:2012 +A2:2016 +A3:2016 +A4:2017 Railway applications - Fixed installations - Electrical safety, earthing and the return circuit
[EN 50124-2]	EN 50124-1: 2017 Railway applications - Insulation coordination – Part 2 Overvoltage and related protection
[EN 50125-3]	EN 50125-3:2003 +AC:2010: Railway applications - Environmental conditions for equipment - Part 3: Equipment for signalling and telecommunications - Part 1: Protective provisions against electrical shock
[EN 50126]	EN 50126-1:2017 – Railway Applications – The Specification and Demonstration of Reliability, Availability, Maintainability and Safety (RAMS)
[EN 50129]	EN 50129:2018 +AC:2019 – Railway applications – Communication, signalling and processing systems – Safety related electronic systems for signalling
[EN 50159]	EN 50159:2010 +A1:2020 – Railway application – Communication, signalling and processing: Safety-related communication in transmission systems
[EN 50310]	EN 50310:2016+A1:2020 - Telecommunications bonding networks for buildings and other structures - 2020-05-03
[EN 50716]	EN 50716:2024 - Railway Applications – Requirements for software development - 2024-10-01
[EN 60529]	IEC 60529:1989/A2:2013/COR1:2019 EN 60529:1991/A2:2013/AC:2019-02 Degrees of protection provided by enclosures (IP Code) (IEC 60529:1989/A2:2013/COR1:2019) (Corrigendum) - 2019-07
[EN 61010-1]	IEC 61010-1:2010+COR:2011+A1:2016 (MOD)+A1:2016/COR1:2019 EN 61010-1:2010 6+A1:2019+A1:2019/AC:2019 Safety requirements for electrical equipment for measurement, control, and laboratory use Part 1: General requirements 2020-04
[EN 60721-3-4]	EN IEC 60721-3-4:2019 Classification of environmental conditions Part 3-4: Classification of groups of environmental parameters and their severities – Stationary use at non-weatherprotected locations (IEC 60721-3-4:2019) 2020-06
[EN 62262]	IEC 62262:2002+AMD1:2021 Degrees of protection provided by enclosures for electrical equipment against external mechanical impacts (IK code) consolidated version 2021-09-03

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[EN 62305]	IEC 62305-1:2024 Protection against lightning – Part 1: General principles 2024-09-12 IEC 62305-2:2024 Protection against lightning – Part 2: Risk management 2024-09-12 IEC 62305-3:2024 Protection against lightning – Part 3: Physical damage to structures and life hazard 2024-09-12 IEC 62305-4:2024 Protection against lightning – Part 4: Electrical and electronic systems within structures 2024-09-12
[EN 62368-1]	EN IEC 62368-1:2025 Audio/video, information and communication technology equipment Part 1: Safety requirements 2025-02-01
[Eu.Doc.20]	EULYNX Generic interface and subsystem requirements, CENELEC Phase: 4, Version: 4.0 (6.A), Approval date: 29.05.2024
[Eu.Doc.7]	EULYNX System definition, CENELEC Phase: 2, Version: 4.2 (2.A), Approval date: 29.05.2024
[Eu.Doc.33]	[Eu.Doc.33], Interface specification SCI-LS, CENELEC Phase: 5, Version: 4.3 (0.A), Approval date: 29.05.2024
[Eu.Doc.37]	[Eu.Doc.37], Signal aspect table, CENELEC Phase: 4-5, Version: 2.2 (1.G), Approval date: 29.05.2024
[Eu.Doc.38]	[Eu.Doc.38], Interface specification SCI-P, CENELEC Phase: 5, Version: 4.2 (1.A), Approval date: 29.05.2024
[Eu.Doc.41]	[Eu.Doc.41], Requirements specification for SCI-ILS, CENELEC Phase: 4, Version: 4.3 (0.A), Approval date: 29.05.2024
[Eu.Doc.42]	[Eu.Doc.42], Interface specification SCI-ILS, CENELEC Phase: 5, Version: 4.3 (0.A), Approval date: 29.05.2024
[Eu.Doc.46]	[Eu.Doc.46], Interface specification SCI-IO, CENELEC Phase: 5, Version: 4.0 (3.A), Approval date: 29.05.2024
[Eu.Doc.47]	[Eu.Doc.47], Requirements specification for SCI-RBC, Cenelec Phase: 4, Version: 4.0 (3.A), Approval date:
[Eu.Doc.48]	[Eu.Doc.48], Interface specification SCI-RBC, Cenelec Phase: 5, Version: 4.0 (1.A), Approval date: 15.06.2023
[Eu.Doc.49]	[Eu.Doc.49], Requirements specification for SCI-CC, CENELEC Phase: 4, Version: 4.2 (1.A), Approval date:
[Eu.Doc.50]	[Eu.Doc.50], Interface specification for SCI-CC, CENELEC Phase: 5, Version: 4.3 (0.A), Approval date: 29.05.2024
[Eu.Doc.76]	[Eu.Doc.76], Interface definition and specification SMI, CENELEC Phase: 5, Version: 2.2 (0.A), Approval date: 29.05.2024
[Eu.Doc.94]	[Eu.Doc.94], Interface specification SDI Generic, CENELEC Phase: 5, Version: 4.2 (0.A), Approval date: 29.05.2024
[Eu.SAT.244]	Speed indication (10 km/h) to
[Eu.SAT.246]	Speed indication (150 km/h)
[Eu.SAT.247]	Speed indication dark - Commanded to dark
[Eu.SAT.346]	Intended dark - Commanded to dark
[Eu.SAT.351]	Not equipped - Signal is not equipped with an extension of the basic signal aspect
[Eu.SAT.352]	Not equipped - Signal is not equipped with a speed indicator
[Eu.SAT.355]	Speed announcement (10 km/h) to
[Eu.SAT.356]	Speed announcement (150 km/h)
[Eu.SAT.357]	Speed announcement dark - Commanded to dark
[Eu.SAT.358]	Not equipped - Signal is not equipped with a speed announcement

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[Eu.SAT.359]	Direction announcements
[ASchG]	ArbeitnehmerInnenschutzgesetz
[EisbAV]	Eisenbahn-ArbeitnehmerInnenschutzverordnung
[AVO Verkehr]	Arbeitnehmerschutzverordnung Verkehr
[AM-VO]	Arbeitsmittelverordnung
[KennV]	Kennzeichnungsverordnung
[Pflichtenheft EBO - Einheitliche Bedienoberfläche]	Einheitliche Bedienoberfläche, EBO2, Bedienkatalog 3BU 15000 4401 DTAPC Ed.03 Einheitliche Bedienoberfläche, EBO2, Meldungskatalog 3BU 15000 4402 DTAPC Ed.03 Einheitliche Bedienoberfläche, EBO2, Symbolkatalog Teil1 3BU 15000 4403 DTAPC Ed.03 Einheitliche Bedienoberfläche, EBO2, Symbolkatalog Teil2 3BU 15000 4404 DTAPC Ed.03 Einheitliche Bedienoberfläche, EBO2, Symbolkatalog Teil3 3BU 15000 4405 DTAPC Ed.03 Einheitliche Bedienoberfläche, EBO2, Symbolkatalog Teil4 3BU 15000 4406 DTAPC Ed.03 Einheitliche Bedienoberfläche, EBO2, Typenkatalog 3BU 15000 4407 DTAPC Ed.02
RAHMENAUSSCHREIBUNG - ETCS STRECKENAUSRÜSTUNG - D.1 Lastenheft]	DSTW Streckenausrüstung ÖBB - Teil D Leistungsbeschreibung und Lastenhefte 3BU 17300 0004 DSAPC Ed. 04
[Schnittstellenbeschreibung X25oIP]	X25-over-IP RaSTA 3BU 15000 2377 DSAPC Ed. 02
[PH X25-IF]	Specification X25 Interface, 3BU 15000 4167 DTAPQ Ed.58
[PH X25 – BFZ Schnittstelle ÖBB]	X25-BFZ – Schnittstelle 3BU 15000 4974 DSAPC Ed.04
[LH – EBO2 / X25]	Lastenheft EBO2 / X25 1.0 01.04.2022

Commented [BD9]: Comment (18) by BAUMGARTNER Dieter (18 Dez 2025 8:39)
Wo ist das zu finden?

Table 1: Reference Documents

1.6 DEFINITIONS, ACRONYMS AND ABBREVIATIONS

See Definitions, Acronyms and Abbreviations [SP_GLOSS] and [EL_GLOSS].

Additional abbreviations not mentioned there: see table below.

TBD	TBD
-----	-----

Table 2: Abbreviations and Acronyms

1.7 STRUCTURE OF THE REQUIREMENTS

See [DIL_RMP] for a description of the SSS requirement structure, the used attributes and their meaning.

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1.8 INFORMATION FOR DOCUMENT GENERATION

The present document was generated by using IBM™ Rational DOORS with following settings:

DOORS/RMF Version:	DOORS 9.7.1.0/RMF 6.1.0.9
DOORS/RMF Module Name:	Helper für EL4 SSS
DOORS/RMF Module Version (Baseline):	EL4 SSS: 1.0
DOORS/RMF Module Path:	/ELEKTRA 4.0/Specification
DOORS/RMF Export View:	WEXP SSS 1258 (DIL)
DOORS/RMF Current Filter:	((Include Object 174) OR (Absolute Number == 173)) OR (Absolute Number == 174)
DOORS/RMF Export Options 1:	2,false,0,0,false,false,0,true,false,
DOORS/RMF Export Options 2:	true,false,true,2,false,false,true,PostprocessAllStyles,\\Thales-AT-Data\Thales-AT\PROJECTS\ReqMng\DOORS\Word Export\DOORS-EXPORT-TEMPLATES\EL4\SRS\Exports\WEXP_1258.log,
DOORS/RMF Export Options 3:	\\Thales-AT-Data\Thales-AT\PROJECTS\ReqMng\DOORS\Word Export\DOORS-EXPORT-TEMPLATES\EL4\SRS\Exports\WEXP_1258.log,true,true,false,false,false,0,true,4,
DOORS/RMF Export Options 4:	3,###,===,true,false,false,false,false,
Documentation-Vob Template Name:	3BU_17300_1258_DTAPA-Template

Commented [PA10]: Comment (28) by PAUL Alexander (19 Dez 2025 10:41)
why not:
TA SRS includes 1258

Table 3: Document Generation Settings

1.9 OPEN POINTS

Update of all figures in accordance with the documents [TECHNICAL_CONCEPTS] and [ARCHITECTURE_DRAWING].

Topic “Interface to DDSP” -> remains as open point for the first DIL baseline (phase 1).

Topic “Geo-Redundancy” -> remains as open point for the first DIL baseline (phase 1).

Topics that require further clarification with the customer.

Review values concerning “Allocation” with architect/design team.

Review values concerning “Verification” with test manager.

Check completeness of attribute “Origin”.

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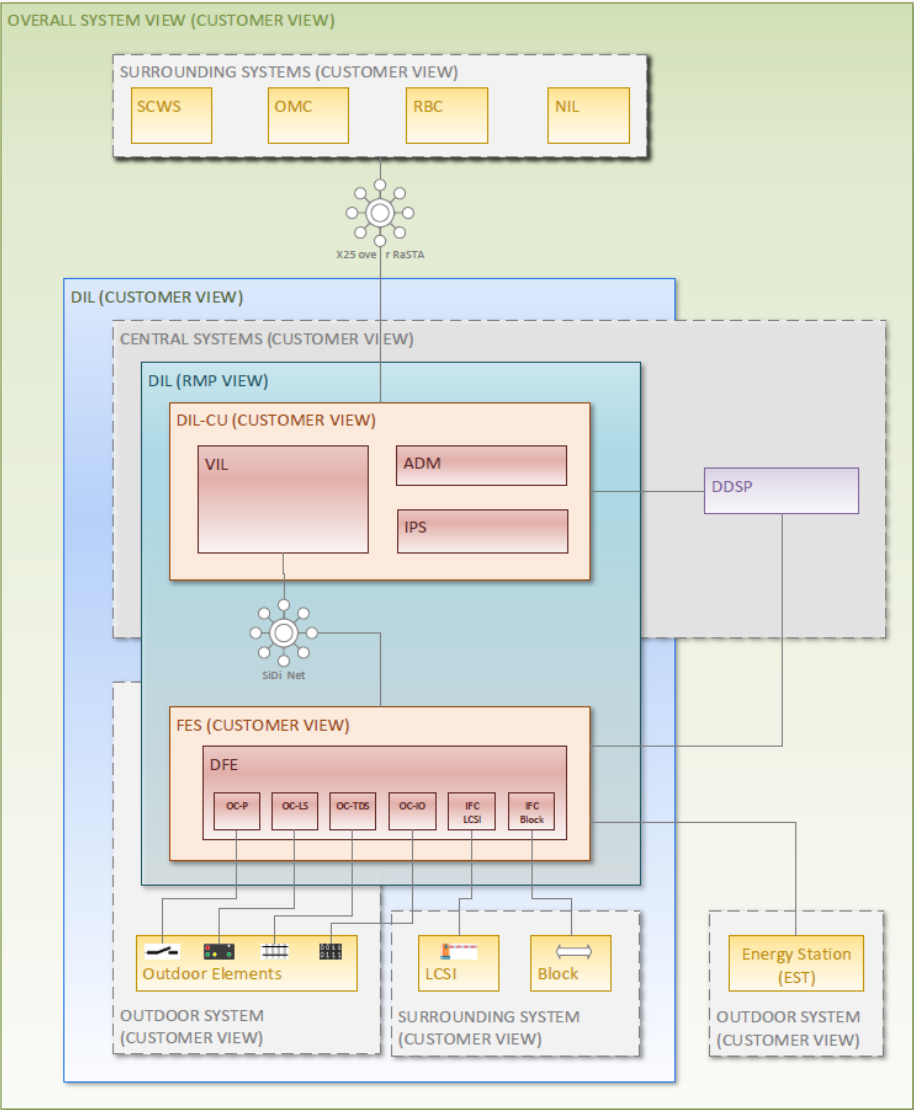
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2. OVERVIEW

The following picture gives an overview of the overall system and the [DIL] system considering both the technical project view and the customer's view.



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Figure 1: System Overview

For an overview and a description of the system please refer to the document [TECHNICAL_CONCEPTS], chapter 1.

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3. REQUIREMENTS

3.1 DATA CENTER

3.1.1 GENERAL REQUIREMENTS & APPLICABLE STANDARDS/REGULATIONS

The [DC] is a unit of

- all of [DIL-CU] ([VIL Instance]s including ~~their~~ [IPS] services and [IPS] hardware),
- network technology,
- air conditioning,
- additional applications ([DDSP]), [UPS], and
- its physical location.

The [DC] is distributed across 2-1 (basic variant) or 4-2 (geo-redundant variant) locations and is specified by the customer.

EL4-DIL-SSS-REQ-01082		Data Center - General Requirements		(Draft)			
Fe-The following requirements from the [SSS_DIL_GENERAL] shall be fulfilled in relation to the [DC]: → Air conditioning → Degree of protection with regard to dust and water resistance (OEVE/OENORM EN 60529) → Fire protection (Eu.Doc.114 - EULYNX Security Specification M00001 - Fire Protection) → Employee protection (AschG) → Environmental protection → Earthing and overvoltage → Environmental conditions							
Comments:							
Category:		GCR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [20_17]	
Target Release:		DIL_OBB_1.0.0		Scope:		System	
Allocation:		DIL		Verification:		< will be added in a later document edition >	
				See also:			
Issues/Decisions:							
LH Text:		1.1.0-2 Anforderungen aus dem <03 Teillastenheft Allgemeine Anforderungen> 1 müssen im Bezug auf das Rechenzentrum eingehalten werden. Dies gilt insbesondere für die Themen: • Klimatisierung • Schutzgrad hinsichtlich Staub- & Wasserfestigkeit(OEVE/OENORM EN 60529) • Brandschutz (Eu.Doc.114 - EULYNX Security Specification M00001 - Fire Protection) • Arbeitnehmerschutz (AschG) • Umweltschutz • Erdung & Überspannung • Umweltbedingungen					
PH Text:		Wird akzeptiert.					

Commented [FM11]: Proposed Change (3) by FRANZL Michael (17 Dez 2025 14:29)
Confusion with 2 x 2oo2 and 4 x 2oo2?

Commented [BR12]: Comment (7) by BUEGER Roman (17 Dez 2025 14:37)
Given only this key words it is not clear which requirements from SS_DIL_GENERAL shall be fulfilled here. E.g for air conditioning or overvoltage I couldn't find any requirements at all.
BAUMGARTNER Dieter (18 Dez 2025 8:44)
Das gilt auch für weiter unten!

Commented [RC13]: Comment (23) by RUGGENTHALER Christoph (18 Dez 2025 15:56)
IPS (HW)

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EL4-DIL-SSS-REQ-01083		Data Center -Security Requirements		(Draft)
The following security requirements shall be met with regard to the [DC]:				
→ [NIS-G] – critical infrastructure (perimeter protection, building integrity, etc.)				
→ System protection				
→ [PH_PART_D2.4]				
→ [DIL_Security_Concept]				
Comments:				
Category:	SER	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_18]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.1.0-3 Security Anforderungen müssen im Bezug auf das Rechenzentrum eingehalten werden. Dies gilt insbesondere für die Themen: • NIS-G - kritische Infrastruktur (Perimeterschutz, Gebäude-Integrität etc.) • Systemschutz • DSTW LH Security • DSTW Konzept Security			
PH Text:	Die Security Anforderungen werden unter dem Aspekt des DSTW-ZE + VSTW berücksichtigt und erfüllt.			

Commented [RC14]: Comment (24) by RUGGENTHALER Christoph (18 Dez 2025 15:57) all

EL4-DIL-SSS-REQ-01084		Data Center - Premises - Applicable Standards and Regulations		(In negotiation)
When providing premises for [DIL-CU] servers and other [DIL] applications, the applicable standards and regulations listed in the reference list shall be observed.				
Comments: == OPEN POINT == Add a complete list here!				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_19]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.1.0-4 Für die Bereitstellung der Räumlichkeiten für DSTW-ZE Server und weitere DSTW-Applikationen sind geltende Normen und Regelwerke laut Referenzliste einzuhalten.			
PH Text:	Wird akzeptiert.			

Commented [BD15]: Comment (19) by BAUMGARTNER Dieter (18 Dez 2025 8:45)
Das ist zu wenig, manche Normen geben mehrere Auswahlmöglichkeiten

3.1.2 DATA CENTER LOCATION

== OPEN POINT ==

To which extent are these requirements in the responsibility of the supplier??

A [DC] location is defined as a building that houses at least the hardware and software components of the [DC] for a [DIL-CU] and its supplementary [DIL] applications (see [DIL] applications) and makes them available for operation in exclusive and enclosed premises.

[DIL-CU] is understood to be the combination of interlocking software ([VIL]) and [COTS]-[IPS] server hardware.

The following points are assumed for calculating the requirements for the [DC]:

Commented [FM16]: Proposed Change (4) by FRANZL Michael (17 Dez 2025 14:32)
E-SHU (server hardware independence) is no longer required by the customer.

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- Following the tender, two contractors will supply [DIL-CU] ~~and [VIL]~~ to ÖBB.
- The total number of [VIL]s that can be equipped is 660
- One [VIL] application requires two [CPU] cores of mid-range [COTS] server hardware
- Each server requires a hypervisor that requires 2 [CPU] cores of mid-range [COTS] server hardware
- Mid-range [COTS] server hardware has 2 [CPU]s with 16 [CPU] cores each
- Mid-range [COTS] server hardware has a height of 2 height units
- Mid-range [COTS] server hardware is equipped with a redundant power supply
- A power supply unit for mid-range [COTS] server hardware can provide up to 1000W of power

TA See also (DXL): ==> []

In the context of these specifications, [DIL] applications are described as software and hardware components that are related to the [DIL] in addition to the operation of [VIL]. This includes at least the following applications:

- Maintenance & Data Management (DDSP)
- Diagnostic tool
- Standard Security Platform (SSP)
- Storage server

EL4-DIL-SSS-REQ-01085		Data Center Location - Geo-Redundancy		(Draft)
At least two [DC] locations shall be provided to achieve geo-redundancy.				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_22]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.2.0-2 Für die Realisierung der Georedundanz müssen mindestens zwei Rechenzentrum Standorte zur Verfügung gestellt werden.			
PH Text:	Wird akzeptiert. Gegebenenfalls ist ein dritter Standort (kein Rechenzentrum) für die Lösung des Split-Brain-Problems erforderlich.			

Commented [FM17]: Proposed Change (5) by FRANZL Michael (17 Dez 2025 14:33)
The second last sentence includes VIL in the CU.

Commented [FM18]: Comment (8) by FRANZL Michael (17 Dez 2025 14:39)
Shouldn't this be VIL redundancy instead of application?

Commented [PA19]: Comment (16) by PAUL Alexander (18 Dez 2025 7:57)
unclear reference

Commented [RC20]: Comment (25) by RUGGENTHALER Christoph (18 Dez 2025 16:05)
all

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EL4-DIL-SSS-REQ-01086		Data Center Location - Distance		(Draft)	
[DC] locations shall be at least 200 km apart.					
Comments:					
Category:	GCR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [20_23]		
Target Release:	DIL_OBB_1.0.0	Scope:	System		
Allocation:	DIL	Verification:	< will be added in a later document edition >		
		See also:			
Issues/Decisions:					
LH Text:	1.2.0-3 Rechenzentrum Standorte müssen mindestens 200km auseinander liegen.				
PH Text:	Wird akzeptiert.				

EL4-DIL-SSS-REQ-01573		Data Center Location - Network Accordance		(Draft)	
The geographical locations of the [DC] shall be selected in accordance with the network concept or network architecture and in line with [DIN EN50600].					
Comments:					
Category:	GCR		Priority:	Optional	
Originator:	Customer		Origin:	==> [EL4 PH] [20_24]	
Target Release:	DIL_OBB_1.0.0		Scope:	System	
Allocation:	DIL		Verification:	< will be added in a later document edition >	
			See also:		
Issues/Decisions:					
LH Text:	1.2.0-4 Die geographischen Standorte des Rechenzentrums sollen in Abstimmung mit dem Netzwerkkonzept bzw. der Netzwerkarchitektur und im Sinne der DIN EN50600 gewählt werden.				
PH Text:					

EL4-DIL-SSS-REQ-01087		Data Center Location - [Eu.Doc.114] and [RW01.10]		(Draft)	
When determining the location for the [DC] building, the requirements of the [EULYNX] specification [Eu.Doc.114] and the [RW01.10] regulations shall be observed in order to avoid disruptions (from major incidents such as prolonged power failures, earthquakes, etc. to disasters and circumstances of force majeure).					
Comments:					
Category:	GCR	Priority:	Mandatory		
Originator:	Customer	Origin:	=> [EL4 PH] [20_25]		
Target Release:	DIL_OBB_1.0.0	Scope:	System		
Allocation:	DIL	Verification:	< will be added in a later document edition >		
		See also:			
Issues/Decisions:					
LH Text:		1.2.0-5 Bei der Bestimmung des Standorts für das Gebäude des Rechenzentrums müssen die Vorgaben der EULYNX-Spezifikation (siehe M00005 [92]) und des Regelwerks RW01.10 eingehalten werden, um Beeinträchtigungen (von Großschadensereignissen wie z.B. längerfristiger Stromausfällen, Erdbeben etc bis hin zu Katastrophenfällen und Umstände höherer Gewalt) zu vermeiden.			
PH Text:		Wird akzeptiert.			

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EL4-DIL-SSS-REQ-01088		Data Center Location - Data Connection		(Draft)
For the operation of the [DIL], each [DC] location shall have a sufficient data connection (via fibre optic cable) via two physically independent cable entries, which are either owned/operated by ÖBB Infrastruktur AG or leased and can be used by ÖBB Infrastruktur.				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_26]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.2.0-6 Für den Betrieb des DSTW muss jeder Rechenzentrum Standort hinsichtlich der Netzwerk-Anbindung erschlossen sein. Das bedeutet, dass dieser Standort über eine ausreichende Datenanbindung (über LWL) über zwei physikalisch voneinander unabhängige Kabeleinführungen verfügen muss, welche entweder im Besitz/Betrieb von ÖBB Infrastruktur AG sind, oder angemietet sind und durch die ÖBB Infrastruktur verwendet werden können.			
PH Text:	Wird akzeptiert.			

EL4-DIL-SSS-REQ-01089		Data Center Location - Power Supply Connection		(Draft)
For the operation of the [DIL], each [DC] location shall have a sufficient connection to the power supply consisting of at least two physically separate connections (path redundancy) and different sources (e.g. providers), which are either owned/operated by ÖBB Infrastruktur AG or leased and can be used by ÖBB Infrastruktur.				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_27]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
See also:				
Issues/Decisions:				
LH Text:	1.2.0-7 Für den Betrieb des DSTW muss jeder Rechenzentrum Standort hinsichtlich der Stromversorgung erschlossen sein. Das bedeutet, dass dieser Standort über eine ausreichende Anbindung aus mindestens zwei physikalisch getrennten Verbindungen (Wegeredundanz) und unterschiedlichen Quellen (z.B. Anbieter) verfügt, welche entweder im Besitz/Betrieb von ÖBB Infrastruktur AG sind, oder angemietet sind und durch die ÖBB Infrastruktur verwendet werden können.			
PH Text:	Wird akzeptiert.			

EL4-DIL-SSS-REQ-01090		Data Center Location - Personnel Restrictions		(Draft)
Personnel from one union may only operate, maintain, inspect or troubleshoot components of their own union (e.g. networks, safety technology, etc.) and shall be trained accordingly for use in the [DC].				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_28]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.2.0-8 Personal eines Gewerks darf nur Komponenten des eigenen Gewerks (z.B.: Netze, Sicherungstechnik, etc.) bedienen, warten, inspizieren oder entstoren und muss entsprechend für den Einsatz im Rechenzentrum geschult sein.			
PH Text:	Wird akzeptiert.			

Commented [BD21]: Comment (20) by BAUMGARTNER Dieter (18 Dez 2025 8:54)
Warum muss man da bei der Gewerkschaft sein?

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EL4-DIL-SSS-REQ-01091		Data Center Location - Space for Doors and Aisles		(Draft)
When calculating the space requirements, it shall be taken into account that the doors (single or double) of the server cabinets or racks can be opened and that appropriate aisles between rows of cabinets shall be planned.				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_31]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
See also:				
Issues/Decisions:				
LH Text:	1.2.0-11 Bei der Berechnung des Platzbedarfs muss mit einberechnet werden, dass die Türen (ein- oder zweiteilig) der Serverschränke bzw. -Racks geöffnet werden können und entsprechende Gänge zwischen Schrankreihen eingeplant werden.			
PH Text:	Die COTS Komponenten (Server- wie Netzwerkkomponenten) sind für die Assemblierung in 19" Racks konzipiert. Die vom AN qualifizierten Server- und Netzwerkschränke haben Abmessungen von 800mm x 1200mm x 2000mm (BxTxH) sowie vorne wie hinten Doppeltüren mit jeweils 400mm Breite.			

Commented [BD22]: Comment (21) by BAUMGARTNER Dieter (18 Dez 2025 8:55)
Wie ist appropriate definiert, das ist so nicht testbar.
Verweis auf unten wäre hilfreich

EL4-DIL-SSS-REQ-01092		Data Center Location - Space between Cabinets		(Draft)
In the [DC], a minimum clearance width of 800 mm shall be maintained between two rows of cabinets.				
Comments:		Reason 1: Escape route Reason 2: Double doors of the cabinets (400mm wide) shall be able to open and close simultaneously on the front and back.		
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_33]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.2.0-13 Zwischen den Schrankreihen muss eine Durchgangsbreite von mindestens 80cm eingehalten werden, um einen Fluchtweg zu ermöglichen.			
PH Text:	Die Anforderung wird unter dem Aspekt des DSTW-ZE + VSTW zur Verfügung gestellten Schränken berücksichtigt und erfüllt. Die Server- und Netzwerkschränke haben Abmessungen von 800mm x 1200mm x 2000mm (BxTxH) sowie vorne wie hinten Doppeltüren mit jeweils 400mm Breite. Dadurch wird ein Öffnen der Türen von gegenüberliegenden Schränken bei Einhaltung der 80cm Durchgangsbreite ermöglicht.			

EL4-DIL-SSS-REQ-01093		Data Center Location - Check of Supply Chain		(Draft)
The entire supply chain for [DC] components (up to the supplier locking system) shall be checked against the Supplier Management Checklist (see M00014 [92]).				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_34]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.2.0-14 Die ganze Supply-Chain für Komponenten des Rechenzentrums (bis Zulieferer Schließsystem) muss hinsichtlich der Supplier Management Checkliste überprüft werden (siehe M00014 [92]).			
PH Text:	Die Anforderung wird unter dem Aspekt des DSTW-ZE + VSTW für die zur Verfügung gestellten Komponenten berücksichtigt und erfüllt.			

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EL4-DIL-SSS-REQ-01094		Data Center Location - Spare Parts Warehouse - (Draft)	
Security Technology			
Due to high availability requirements and in the event of an operational failure, a spare parts warehouse for security-safety technology components shall be planned so that the IH-maintainer has the LS-light signal and network components available as quickly as possible.			
→ A room shall be provided for this purpose and equipped with a heavy-duty control system.			
→ A room separate from the computer rooms shall be provided for the spare parts warehouse.			
→ The size of the space shall be determined in consultation with the responsible operations management and based on the service and maintenance contract (see also [M00033]).			
Comments:			
Category:	GCR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [20_35] ==> [EL4 PH] [20_102]
Target Release:	DIL_OBB_1.0.0	Scope:	System
Allocation:	DIL	Verification:	< will be added in a later document edition >
		See also:	
Issues/Decisions:			
LH Text:		1.2.0-15 Wegen hoher Anforderungen an die Verfügbarkeit und für den Fall eines betrieblichen Ausfalls, muss ein Ersatzteillager für die Komponenten der Sicherungstechnik geplant werden, damit die IH möglichst schnell die LS- und Netzwerk-Komponenten zur Verfügung haben. Dafür ist eine Räumlichkeit vorzusehen und mit einem Schwerlast-Regelsystem auszustatten. Für das Ersatzteillager muss ein von den Rechnerräumlichkeiten separater Raum zur Verfügung gestellt werden. Die Größe der Räumlichkeit muss in Absprache mit der zuständigen Betriebsführung und auf Basis des Service- & Wartungsvertrags festgelegt werden. Siehe auch M00033 [92].	
		1.2.0-17 Grafik Anzahl Racks	
PH Text:		Wird akzeptiert.	

Commented [PA23]: Proposed Change (32) by PAUL Alexander (19 Dez 2025 11:34)
I assume that in this case the interlocking equipment is meant.

Commented [PA24]: Proposed Change (30) by PAUL Alexander (19 Dez 2025 11:30)
better readability.
PAUL Alexander (19 Dez 2025 12:33)
already changed in DOORS.

EL4-DIL-SSS-REQ-01095		Data Center Location - Spare Parts Warehouse - (Draft)	
Network and Transmission Technology			
Due to high availability requirements and in the event of an operational failure, a separate spare parts warehouse shall be planned for the network and transmission technology components so that the IH-maintainer has the LS-light signal and network components available as quickly as possible. A room shall be provided for this purpose and equipped with a heavy-duty control system. A room separate from the computer rooms shall be provided for the spare parts warehouse. The size of the room shall be determined in consultation with the responsible operations management and on the basis of the service and maintenance contract (ee also [M00033]).			
Comments:			
Category:	GCR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [20_36] ==> [EL4 PH] [20_102]
Target Release:	DIL_OBB_1.0.0	Scope:	System
Allocation:	DIL	Verification:	< will be added in a later document edition >
		See also:	
Issues/Decisions:			
LH Text:	1.2.0-16 Wegen hoher Anforderungen an die Verfügbarkeit und für den Fall eines betrieblichen Ausfall muss ein separates Ersatzteillager für die Komponenten der Netzwerk- & Übertragungstechnik geplant werden, damit die IH möglichst schnell die LS- und Netzwerk-Komponenten zur Verfügung haben. Dafür ist eine Räumlichkeit vorzusehen und mit einem Schwerlast-Regelsystem auszustatten. Für das Ersatzteillager muss ein von den Rechnerräumlichkeiten separater Raum zur Verfügung gestellt werden. Die Größe der Räumlichkeit muss in Absprache mit der zuständigen Betriebsführung und auf Basis des Service- & Wartungsvertrags festgelegt werden. Siehe auch M00033 [92]. 1.2.0-17 Grafik Anzahl Racks		
PH Text:	Wird akzeptiert.		

Commented [PA25]: Proposed Change (31) by PAUL Alexander (19 Dez 2025 11:32)

PAUL Alexander (19 Dez 2025 12:33)
already changed in DOORS.

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EL4-DIL-SSS-REQ-01096		Data Center Location - Space for 19" Servers		(Draft)
Space shall be provided for the required 19" servers for the operation of the [DIL-CU]. In total, space shall be provided for at least				
→ 12 racks with 42 U per AN-contractor [DIL-CU] (including 4 racks for redundancy)				
→ 6 racks with 42 U per AN-contractor [DIL-CU] for conversion and migration phases shall be provided.				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_38]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
See also:				
Issues/Decisions:				
LH Text:	1.2.0-18 Für den Betrieb der DSTW-ZE muss Platz für benötigte Server in 19"-Bauform zur Verfügung gestellt werden. Insgesamt ist Platz für mindestens			
	• 12 Racks à 42 HE pro AN DSTW-ZE (davon 4 Racks für Redundanz)			
	• 6 Racks à 42 HE pro AN DSTW-ZE für Umbau- & Migrationsphasen vorzusehen.			
PH Text:	Wird akzeptiert.			

Commented [PA26]: Proposed Change (34) by PAUL Alexander (19 Dez 2025 11:37)

PAUL Alexander (19 Dez 2025 12:32)
already changed in DOORS.

Commented [PA27]: Proposed Change (33) by PAUL Alexander (19 Dez 2025 11:36)

PAUL Alexander (19 Dez 2025 12:32)
already changed in DOORS.

EL4-DIL-SSS-REQ-01097		Data Center Location - Space for Network and Transmission Technology		(Draft)
Space shall be provided for the necessary network and transmission technology to operate the [DIL-CU]. In total, space shall be provided for at least				
→ 4 network technology cabinets				
→ 2 fibre optic cabinets				
→ 4 transmission technology cabinets				
→ 2 spare cabinets				
.				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_39]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.2.0-19 Für den Betrieb der DSTW-ZE muss Platz für benötigte Netzwerk- & Übertragungstechnik zur Verfügung gestellt werden. Insgesamt ist Platz für mindestens			
	• 4 Netzwerktechnik-Schränke			
	• 2 LWL-Schränke			
	• 4 Übertragungstechnik-Schränke			
	• 2 Reserve-Schränke vorzusehen.			
PH Text:	Wird akzeptiert.			

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EL4-DIL-SSS-REQ-01098		Data Center Location - Space for [DIL] Applications		(Draft)
Space shall be provided for [DIL] applications in the [DC].				
Space shall be provided for at least				
→ 2 racks in 19" design, each with 42 U for storage servers				
→ 2 racks in 19" design, each with 42 U, for the Standard Security Platform (SSP)				
→ 2 racks in 19" design, each with 42 U, for maintenance & data management (DDSP)				
→ 5 racks in 19" design, each with 42 U, as a reserve				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_40]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.2.0-20 Es ist Platz für DSTW-Applikationen des Rechenzentrums vorzusehen. Hierbei ist Platz für mindestens			
	• 2 Racks in 19"-Bauweise à 42 HE für Storage Server			
	• 2 Racks in 19"-Bauweise à 42 HE für die Standard Security Plattform (SSP)			
	• 2 Racks in 19"-Bauweise à 42 HE für das Maintenance & Data Management (DDSP)			
	• 5 Racks in 19"-Bauweise à 42 HE als Reserve vorzusehen.			
PH Text:	Wird akzeptiert.			

EL4-DIL-SSS-REQ-01099		Data Center Location - Space for Operating Stations		(Draft)
One location of the [DC] shall provide space for the following operating stations outside the computer rooms: → Local DDSP operating stations → Test workstations The number of workstations required shall be specified during the construction of the [DC].				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_41]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.2.0-21 Ein Standort des Rechenzentrums muss Flächen für folgende Bedienplätze außerhalb der Rechnerräume zur Verfügung stellen: • Lokale Bedienplätze DDSP • Testarbeitsplätze Die Anzahl der geforderten Plätze müssen während der Errichtung des Rechenzentrums genannt werden.			
PH Text:	Wird akzeptiert.			

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EL4-DIL-SSS-REQ-01100		Personnel & Employee Protection - Requirements for Staff		(Draft)
Until the [DC] is constructed, a concept shall be developed that covers the personnel requirements and necessary needs of the on-site staff.				
Applicable laws, standards, regulations and rules shall be observed.				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_43]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
See also:				
Issues/Decisions:				
LH Text:	1.3.0-1 Bis zur Errichtung des Rechenzentrums muss ein Konzept ausgearbeitet werden, welches den Personalbedarf, sowie notwendige Bedürfnisse des Personals vor Ort deckt. Dabei müssen geltende Gesetze, Normen, Verordnungen und Regelwerke beachtet werden.			
PH Text:	Wird akzeptiert.			

Commented [BD28]: Comment (22) by BAUMGARTNER Dieter (18 Dez 2025 8:57)
das ist aber sehr schwammig

EL4-DIL-SSS-REQ-01101		Personnel & Employee Protection - Staff Location		(Draft)
The operating personnel shall be located within the [DC] site in such a way that the service levels promised by the customer can be maintained.				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_44]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.3.0-2 Das Bedienpersonal muss innerhalb des Rechenzentrum-Standorts so lokalisiert sein, dass die zugesagten Service-Levels des AG eingehalten werden können.			
PH Text:	Wird akzeptiert.			

Commented [PA29]: Comment (35) by PAUL Alexander (19 Dez 2025 11:45)
Formulation in German is also not fully clear but I assume that in this case the service levels are promised by the *contractor* in the context of this tender?

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EL4-DIL-SSS-REQ-01102		Personnel & Employee Protection - Local Workplace		(Draft)
A local workplace at the [DC] location shall provide at least the following: → A floor space in accordance with employee protection regulations → Work furniture (desk, chair) in accordance with employee protection (ergonomic workplace) → At least 5 network connections to the Datennetz2020 data network, including the necessary network components (switch, router) → In the case of DDSP: connection to the [SiDiNet]				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_45]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.3.0-3 Ein lokaler Arbeitsplatz des Rechenzentrum Standorts muss mindestens zur Verfügung stellen: • Eine Grundfläche entsprechend dem Arbeitnehmerschutz • Arbeitsmöbel (Schreibtisch, Sessel) entsprechend dem Arbeitnehmerschutz (Ergonomischer Arbeitsplatz) • Mindestens 5 Netzwerkanschlüsse an das Datennetz2020 inklusive notwendiger Netzwerkkomponenten (Switch, Router) • Im Falle des DDSP: Anschluss an das SiDiNet			
PH Text:	Wird akzeptiert.			

EL4-DIL-SSS-REQ-01103		Personnel & Employee Protection - Design for Permanent Staffing		(Draft)
The computer rooms and control stations of the [DIL] shall be designed for permanent staffing of the imaintenance-maintenance personnel				
Comments:		The number of permanently present maintenance and operations management personnel can be assumed to be on a 24/7 rotation (including holiday and sick leave reserves).		
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_46]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
See also:				
Issues/Decisions:				
LH Text:	1.3.0-4 Es muss vorgesehen werden, dass im Bereich der Rechnerräume und Bedienplätze des Instandhaltungspersonal des DSTW dauerhaft Personal eingesetzt wird. Die Anzahl des dauerhaft anwesenden Instandhaltungs- & Betriebsführungspersonal kann als Turnus mit 24/7-Anwesenheit angenommen werden (inklusive Urlaubs- & Krankenstandsreserve).			
PH Text:	Wird akzeptiert.			

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EL4-DIL-SSS-REQ-01104 Personnel & Employee Protection - Deployment (Draft) of Shift Personnel.			
In accordance with applicable laws and employee protection regulations, appropriate precautions shall be taken to enable the deployment of shift personnel. This includes appropriate cloakrooms, social rooms, coffee kitchens, and WC-toilets and washrooms.			
Comments:			
Category:	GCR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [20_47]
Target Release:	DIL_OBB_1.0.0	Scope:	System
Allocation:	DIL	Verification:	< will be added in a later document edition >
See also:			
Issues/Decisions:			
LH Text:	1.3.0-5 Entsprechend geltender Gesetze & Arbeitnehmerschutz sind entsprechende Vorkehrungen zu treffen, damit ein Einsatz von Turnus-Personal möglich ist. Dies schließt entsprechende Garderoben, Sozialräume, Kaffeeküchen sowie WC- & Waschräume mit ein.		
PH Text:	Wird akzeptiert. Seitens AN ist hier keine Mitwirkung erkennbar.		

Commented [PA30]: Proposed Change (36) by PAUL Alexander (19 Dez 2025 11:48)

PAUL Alexander (19 Dez 2025 12:32)
already changed in DOORS.

EL4-DIL-SSS-REQ-01105 Physical Security - Concept (Draft)			
For the implementation of physical security, a concept shall be developed that complies with the relevant physical security requirements, such as the security concept, the [SSS_DIL_GENERAL], valid procedural instructions & the [NIS-G], as well as the requirements of the [EULYNX] specification (see M00002 [92]).			
Comments:			
Category:	SER	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [20_49]
Target Release:	DIL_OBB_1.0.0	Scope:	System
Allocation:	DIL	Verification:	< will be added in a later document edition >
See also:			
Issues/Decisions:			
LH Text:	1.4.0-1 Für die Umsetzung der Physischen Sicherheit ist ein Konzept zu entwickeln, welches den relevanten Vorgaben der Physischen Sicherheit, wie dem Konzept Security, dem <03 Teillastenheft Allgemeine Anforderungen> 1, gültigen Verfahrensanweisungen & dem NIS-G, sowie den Vorgaben der EULYNX-Spezifikation (siehe M00002 [92]) entspricht.		
PH Text:	Wird akzeptiert.		

EL4-DIL-SSS-REQ-01106 Physical Security - Applicable Regulations and (Draft) Procedural Instructions			
To ensure physical security, applicable regulations and procedural instructions shall be followed.			
Comments:			
Category:	SER	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [20_51]
Target Release:	DIL_OBB_1.0.0	Scope:	System
Allocation:	DIL	Verification:	< will be added in a later document edition >
See also:			
Issues/Decisions:			
LH Text:	1.4.0-3 Für die Einhaltung der Physischen Sicherheit müssen geltende Regelwerke und Verfahrensanweisungen eingehalten werden.		
PH Text:	Wird akzeptiert.		

Commented [PA31]: Comment (37) by PAUL Alexander (19 Dez 2025 11:49)
could be also safety in this context, unclear from German text.

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EL4-DIL-SSS-REQ-01107Physical Security - Access to Computer Rooms (Draft) - Normal Situation			
Access to computer rooms at the [DC] location shall be controlled electronically and configured individually (e.g. via an authorisation card). In any case, access control shall be [NIS-G] compliant.			
Comments:			
Category:	SER	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [20_52]
Target Release:	DIL_OBB_1.0.0	Scope:	System
Allocation:	DIL	Verification:	< will be added in a later document edition >
See also:			
Issues/Decisions:			
LH Text:	1.4.0-4 Der Zugang zu Rechnerräumen des Rechenzentrum Standorts muss elektronisch geregelt und individuell (z.B. über den Berechtigungsausweis) konfiguriert werden können. In jedem Fall muss die Zugangskontrolle NIS-G konform erfolgen.		
PH Text:	Wird akzeptiert.		

EL4-DIL-SSS-REQ-01108Physical Security - Access to Computer Rooms (Draft) -Malfunction			
In the event of a malfunction (e.g. failure of the access control system due to power failure), it shall be possible to access the computer rooms using a key. This key shall be unique (no other purpose) and kept secure (e.g. in a safe) for the computer rooms at the [DC] location.			
Comments:			
Category:	SER	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [20_53]
Target Release:	DIL_OBB_1.0.0	Scope:	System
Allocation:	DIL	Verification:	< will be added in a later document edition >
See also:			
Issues/Decisions:			
LH Text:	1.4.0-5 Im Störfall (z.B. Ausfall der Zugangskontrolle durch Stromversorgung) muss der Zugang zu den Rechnerräumen mittels eines Schlüssels möglich sein. Dieser Schlüssel muss einzigartig (kein anderer Zweck) für die Rechnerräume des Rechenzentrum-Standorts sicher verwahrt werden (z.B.: über Safe).		
PH Text:	Wird akzeptiert.		

EL4-DIL-SSS-REQ-01109Physical Security - Anti-theft Measures (Draft)			
Anti-theft measures in accordance with EULYNX specifications (see M00032 [92]) shall be considered and implemented in the computer centre.			
Comments:			
Category:	SER	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [20_54]
Target Release:	DIL_OBB_1.0.0	Scope:	System
Allocation:	DIL	Verification:	< will be added in a later document edition >
See also:			
Issues/Decisions:			
LH Text:	1.4.0-6 Es müssen Anti-Diebstahl-Vorkehrungen entsprechend der EULYNX-Vorgaben (siehe M00032 [92]) im Rechenzentrum betrachtet und umgesetzt werden.		
PH Text:	Wird akzeptiert.		

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EL4-DIL-SSS-REQ-01110 Physical Security - Spare Parts Warehouse (Draft)	
All rules regarding physical security shall be applied to the spare parts warehouse in the same way as for the computer room (AR-RZ room).	
Comments:	
Category: SER	Priority: Mandatory
Originator: Customer	Origin: ==> [EL4 PH] [20_55]
Target Release: DIL_OBB_1.0.0	Scope: System
Allocation: DIL	Verification: < will be added in a later document edition >
See also:	
Issues/Decisions:	
LH Text:	1.4.0-7 Für das Ersatzteillager müssen alle Regeln hinsichtlich der Physischen Sicherheit angewendet werden, wie für den Rechnerraum (AR-RZ-Raum).
PH Text:	Wird akzeptiert.

EL4-DIL-SSS-REQ-01111 Physical Security - Fire Protection Concepts (Draft)	
Fire protection concepts shall be developed for the computer centre locations. The following points and topics shall be taken into account: → Oxygen reduction in computer rooms → Structural and technical fire protection in accordance with official requirements → Spare parts storage → Ventilation/air conditioning → Escape routes → Fire compartments (including division of computer rooms)	
Comments:	
Category: SER	Priority: Mandatory
Originator: Customer	Origin: ==> [EL4 PH] [20_56]
Target Release: DIL_OBB_1.0.0	Scope: System
Allocation: DIL	Verification: < will be added in a later document edition >
See also:	
Issues/Decisions:	
LH Text:	1.4.0-8 Für die Standorte des Rechenzentrums müssen Konzepte für den Brandschutz erarbeitet werden. Dabei müssen folgende Punkte beziehungsweise Themen beachtet werden: • Sauerstoffabsenkung in Rechnerräumen • baulicher und technischer Brandschutz entsprechend behördlicher Vorgaben • Ersatzteilhaltung • Belüftung/Klimatisierung • Fluchtwege • Brandabschnitte (inklusive Aufteilung der Rechnerräume)
PH Text:	Wird akzeptiert.

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EL4-DIL-SSS-REQ-01112		Energy Concept - Minimum Power Supply		(Draft)
To operate a [DC] location, a minimum power supply of 200 kW three-phase 400V AC (single-phase 230V AC) 50Hz shall be available without interruption to supply the [DIL-CU], network and transmission technology, and other [DIL] applications.				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_58]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.5.0-1 Für den Betrieb eines Rechenzentrum Standorts muss mindestens eine Leistung von 200 kW dreiphasig 400V AC (einphasig 230V AC) 50Hz für die Versorgung der DSTW-ZE, Netzwerk- & Übertragungstechnik, sowie weitere Applikationen des DSTW unterbrechungsfrei zur Verfügung stellen.			
PH Text:	Wird akzeptiert.			

EL4-DIL-SSS-REQ-01113		Energy Concept - Main [UPS] Distributors		(Draft)
The location of the main [UPS] distributors shall take into account a maximum cable length of 50 metres to the computer cabinets.				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_59]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.5.0-2 Die Situierung der USV-Hauptverteiler muss eine verlegte Kabellänge zu den Rechnerschränken von maximal 50 Metern berücksichtigen.			
PH Text:	Wird akzeptiert.			

EL4-DIL-SSS-REQ-01114		Energy Concept - Minimum Power		(Draft)
For the operation of a [DC] location, a minimum power of 7 kW, 48V DC shall be available to supply the network and transmission technology without interruption.				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_60]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.5.0-3 Für den Betrieb eines Rechenzentrum Standorts muss mindestens eine Leistung von 7 kW, 48V DC für die Versorgung der Netzwerk- & Übertragungstechnik unterbrechungsfrei zur Verfügung stellen.			
PH Text:	Wird akzeptiert.			

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EL4-DIL-SSS-REQ-01115 Energy Concept - Redundant Power Supply (Draft)	
Blackout scenarios: a highly available power supply shall ensure the operation of the [DC] for a minimum of 6 hours after a power failure.	
Comments: Redundant power supply (e.g. from overhead lines or local suppliers)	
Category: GCR	Priority: Mandatory
Originator: Customer	Origin: ==> [EL4 PH] [20_61]
Target Release: DIL_OBB_1.0.0	Scope: System
Allocation: DIL	Verification: < will be added in a later document edition >
See also:	
Issues/Decisions:	
LH Text:	1.5.0-4 Blackout-Szenarien: hochverfügbare Stromversorgung muss den Betrieb des RZ für eine Mindestdauer von 6 Stunden nach Netzausfall gewährleisten. Redundante Stromversorgung (z.B.: aus Oberleitung oder örtlicher Versorger)
PH Text:	Wird akzeptiert. Seitens AN ist hier keine Mitwirkung erkennbar.

EL4-DIL-SSS-REQ-01116 Energy Concept - [UPS] Requirements Specification (Draft)	
When implementing the power supply for a [DC] location, the specifications of the [UPS] requirements specification shall be observed.	
Comments:	
Category: GCR	Priority: Mandatory
Originator: Customer	Origin: ==> [EL4 PH] [20_62]
Target Release: DIL_OBB_1.0.0	Scope: System
Allocation: DIL	Verification: < will be added in a later document edition >
See also:	
Issues/Decisions:	
LH Text:	1.5.0-5 Bei der Umsetzung der Stromversorgung eines Rechenzentrum Standorts müssen Vorgaben des USV-Lastenhefts eingehalten werden.
PH Text:	Wird akzeptiert.

EL4-DIL-SSS-REQ-01117 Energy Concept - Compliance with Availability Values (Draft)	
The availability values assigned to the power supply [SSS_DIL_RAMs] shall be complied with.	
Comments:	
Category: GCR	Priority: Mandatory
Originator: Customer	Origin: ==> [EL4 PH] [20_63]
Target Release: DIL_OBB_1.0.0	Scope: System
Allocation: DIL	Verification: < will be added in a later document edition >
See also:	
Issues/Decisions:	
LH Text:	1.5.0-6 Die an die Stromversorgung zugewiesenen Verfügbarkeitswerte (siehe <02 Teillastenheft RAMS> 1) müssen eingehalten werden.
PH Text:	Wird akzeptiert.

Commented [MF32]: Comment (41) by MOSER Florian (23 Dez 2025 10:57)
nicht in den Referenced Documents vorhanden

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EL4-DIL-SSS-REQ-01118 Energy Concept - Independently Secured Power (Draft) Circuits			
Redundant power supplies for hardware at the [DC] location shall be supplied by independently secured power circuits.			
Comments:			
Category:	GCR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [20_64]
Target Release:	DIL_OBB_1.0.0	Scope:	System
Allocation:	DIL	Verification:	< will be added in a later document edition >
See also:			
Issues/Decisions:			
LH Text:	1.5.0-7 Redundante Netzteile von Hardware des Rechenzentrum Standorts müssen von unabhängig voneinander abgesicherten Stromkreisen versorgt werden.		
PH Text:	Wird akzeptiert.		

EL4-DIL-SSS-REQ-01119 Cooling Concept - Topics (Draft)			
A cooling and air conditioning concept for the [DC] shall be presented. In addition to the other requirements described, the following topics shall be covered: → Removal of waste heat → Achievement of air flow → Floor condition, including necessary air outlets → Use of mesh doors → Definition of cold and warm sides of the servers (fan flow direction)			
Comments:			
Category:	GCR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [20_66]
Target Release:	DIL_OBB_1.0.0	Scope:	System
Allocation:	DIL	Verification:	< will be added in a later document edition >
See also:			
Issues/Decisions:			
LH Text:	1.6.0-1 Es muss ein Konzept Kühlung & Klimatisierung für das Rechenzentrum vorgelegt werden. Dabei müssen neben der weiteren beschriebenen Anforderungen folgende Themen abgedeckt werden: • Abtransport der Abwärme • Erreichung Luftstrom • Beschaffenheit des Bodens inklusive notwendiger Luftauslässe • Einsatz von Gittertüren • Definition von Kalter und Warmer Seite der Server (Strömungsrichtung Lüfter)		
PH Text:	Die Klimatisierung wird vom AG zur Verfügung gestellt. Die Anforderungen werden unter dem Aspekt des DSTW-ZE + VSTW erfüllt. Die Strömungsrichtung der eingesetzten Komponenten verläuft von vorne nach hinten. Für einen ausreichend kühlenden Luftstrom sorgen Luftauslässe aus den Bodenplatten vor den Schränken, des Weiteren sind die Schranktüren als Gittertüren für ausreichend Luftdurchlass ausgestattet. Für die Leistungsaufnahme siehe Anforderung 20_58. Bei der Situierung der Schränke muss darauf geachtet werden, dass die Schränke nicht die warme Abluft der Vorderschränke zu deren Kühlung ansaugen.		

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EL4-DIL-SSS-REQ-01120		Cooling Concept - Air Conditioning		(Draft)
Compliance with the required temperatures shall be achieved by means of air conditioning for all computer rooms.				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_67]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.6.0-2 Die Einhaltung der geforderten Temperaturen ist durch eine Klimaanlage für alle Rechnerräume zu erreichen.			
PH Text:	Wird akzeptiert.			

EL4-DIL-SSS-REQ-01121		Cooling Concept - Conformity with EULYNX		(Draft)
The requirements of the [EULYNX] specification for air conditioning, in particular the availability of the air conditioning system, shall be met (see EU.SecSpec.1577).				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_68]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.6.0-3 Die Anforderungen gemäß EULYNX-Spezifikation an die Klimatisierung, insbesondere an die Verfügbarkeit der Klimaanlage, müssen eingehalten werden (siehe EU.SecSpec.1577).			
PH Text:	Wird akzeptiert.			

EL4-DIL-SSS-REQ-01122		Cooling Concept - Air Flow		(Draft)
The air flow through the computer rooms shall be achieved by air conditioning in accordance with the general requirements.				
Comments:		The conditions of the premises can be used for this purpose (e.g. air conditioning of the server racks directly through the raised floor).		
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_69]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.6.0-4 Der Luftstrom durch die Rechnerräume muss entsprechend der Allgemeinen Anforderungen durch die Klimatisierung erreicht werden. Die Gegebenheiten der Räumlichkeiten können hierfür herangezogen werden (z.B.: Klimatisierung der Server-Racks direkt durch den Doppelboden).			
PH Text:	Die Klimatisierung wird vom AG zur Verfügung gestellt. Die Anforderungen aus dem Teillastenheft Allgemeine Anforderungen werden unter dem Aspekt des DSTW-ZE + VSTW erfüllt. Die Strömungsrichtung der eingesetzten Komponenten verläuft von vorne nach hinten. Für einen ausreichend kühlenden Luftstrom sorgen Luftauslässe aus den Bodenplatten vor den Schränken, des Weiteren sind die Schranktüren als Gittertüren für ausreichend Luftdurchlass ausgestattet. Für die Leistungsaufnahme siehe Anforderung 20_58.			

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EL4-DIL-SSS-REQ-01123		Cooling Concept - Regulation of Cooling Capacity		(Draft)
It shall be possible to regulate the cooling capacity of the air conditioning system (staggered activation of capacity).				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_70]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.6.0-5 Die Entwärmungsleistung der Klimaanlage muss reguliert werden können (gestaffelte Aufschaltung der Leistung).			
PH Text:	Wird akzeptiert.			

EL4-DIL-SSS-REQ-01124		Cooling Concept - Redundancy Concept for Cable Routing.		(Draft)
A redundancy concept shall be developed for cable routing in the raised floor and ceiling, including supply lines (power supply, network) and cabling within the [DC].				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_71]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.6.0-6 Für die Kabelführung muss ein Redundanzkonzept erarbeitet werden. Dabei müssen Verkabelungswege im Doppelboden und in der Decke mitbetrachtet werden. Dies schließt Zuleitungen (Stromversorgung, Netzwerk) und Verkabelung innerhalb des Rechenzentrums mit ein.			
PH Text:	Wird akzeptiert.			

EL4-DIL-SSS-REQ-01125		Cooling Concept - Raised Floors		(Draft)
Antistatic raised floors shall be used in the computer rooms.				
The raised floor shall be of sufficient height to ensure cooling despite the cabling in the raised floor (e.g. stand height of approx. 50–60 cm).				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_72]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.6.0-7 Im Bereich der Rechnerräume ist der Einsatz von antistatischen Doppelböden vorzusehen. muss der Doppelboden eine ausreichende Höhe aufweisen, um die Kühlung trotz Verkabelung im Doppelboden zu gewährleisten (z.B. Ständerhöhe von ca. 50 - 60 cm).			
PH Text:	Wird akzeptiert.			

Commented [FM33]: Comment (11) by FRANZL Michael (17 Dez 2025 14:57)
Because this requirement mentions cable routing, here is a SP-SEC requirement that should not be forgotten because it can trigger a requirement on the DC: SP-SEC-Pgrm-5.2-2 requires physical protection of cables to mitigate risks in case data at transit is not protected in regards to confidentiality or integrity.

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EL4-DIL-SSS-REQ-01126 Cooling Concept - Floor Panels (Draft)	
Floor panels with air outlets and a free cross-section of at least 20% shall be provided between the rows of cabinets to allow cold air to be drawn in from the front of the IT cabinets.	
Comments:	
Category: GCR Originator: Customer Target Release: DIL_OBB_1.0.0 Allocation: DIL	Priority: Mandatory Origin: ==> [EL4 PH] [20_73] Scope: System Verification: < will be added in a later document edition > See also:
Issues/Decisions:	
LH Text:	1.6.0-8 Zwischen den Schrankreihen müssen Bodenplatten mit Luftauslässen und einen freien Querschnitt von mindestens 20 % vorgesehen werden, um den IT-Schränken die Kaltluft-Ansaugung von vorne zu ermöglichen.
PH Text:	Wird akzeptiert.

EL4-DIL-SSS-REQ-01127 Cooling Concept - Compliance of Temperatures (Draft) in Computer Room	
The temperatures in the computer rooms of the [DC] location shall comply with the general requirements and correspond to the category "air-conditioned building".	
Comments:	
Category: GCR Originator: Customer Target Release: DIL_OBB_1.0.0 Allocation: DIL	Priority: Mandatory Origin: ==> [EL4 PH] [20_74] Scope: System Verification: < will be added in a later document edition > See also:
Issues/Decisions:	
LH Text:	1.6.0-9 Die Temperaturen in den Rechnerräumen des Rechenzentrum Standorts müssen gemäß Allgemeinen Anforderungen eingehalten werden und der Kategorie "Gebäude klimatisiert" entsprechen.
PH Text:	Wird akzeptiert.

EL4-DIL-SSS-REQ-01129 Server Hardware Requirements - Top-of-the-rack (Draft) Switches	
Top-of-the-rack switches shall be installed in the server cabinets for server cabling. The number of switches (ports) will be determined during the clustering planning phase.	
Comments:	
Category: GCR Originator: Customer Target Release: DIL_OBB_1.0.0 Allocation: DIL	Priority: Mandatory Origin: ==> [EL4 PH] [20_78] Scope: System Verification: < will be added in a later document edition > See also:
Issues/Decisions:	
LH Text:	1.7.0-3 In den Server-Schränken müssen Top-of-the-Rack Switches für die Verkabelung der Server ausgeführt werden. Die Anzahl der Switches (Ports) werden im Rahmen der Planung der Clusterung bestimmt.
PH Text:	Wird akzeptiert.

Commented [RC34]: Comment (26) by RUGGENTHALER Christoph (18 Dez 2025 16:15)
IPS (HW)

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EL4-DIL-SSS-REQ-01131		Server Hardware Requirements - Minimum Installation Depth		(Draft)
The racks shall provide a minimum installation depth of 100 cm for servers and network technology components.				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_80]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.7.0-5 Die Racks müssen in der Tiefe mindestens 100cm Einbautiefe für Server und Netzwerktechnikkomponenten zur Verfügung stellen.			
PH Text:	Wird akzeptiert.			

EL4-DIL-SSS-REQ-01132		Server Hardware Requirements - Power Supply - PDUs		(Draft)
The power supply for the server hardware shall be implemented in the rack using vertically mounted [PDU]s.				
Comments:		See also "Schematic diagram of [DIL-CU] server cabinet".		
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_81]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
See also:				
Issues/Decisions:				
LH Text:	1.7.0-6 Die Stromversorgung der Server Hardware muss im Rack vertikal montierte PDU (Power Distribution Unit) realisiert werden. Die Konfiguration der Steckverbinder und Kabel muss in Abstimmung mit dem Lieferanten der Server Hardware durchgeführt werden. Siehe auch "Schematische Darstellung DSTW-ZE Serverschrank"			
PH Text:	Wird akzeptiert.			

EL4-DIL-SSS-REQ-01574		Server Hardware Requirements - Power Supply - (Draft)	
Configuration of Connectors and Cables			
The power supply for the server hardware shall be implemented in the rack using vertically mounted [PDU]s.			
The configuration of the connectors and cables shall be carried out in consultation with the supplier of the server hardware.			
Comments:		See also "Schematic diagram of [DIL-CU] server cabinet".	
Category:	GCR	Priority:	Mandatory
Originator:	Customer	Origin:	=> [EL4 PH] [20_81]
Target Release:	DIL_OBB_1.0.0	Scope:	System
Allocation:	DIL	Verification:	< will be added in a later document edition >
		See also:	
Issues/Decisions:			
LH Text:	1.7.0-6 Die Stromversorgung der Server Hardware muss im Rack vertikal montierte PDU (Power Distribution Unit) realisiert werden. Die Konfiguration der Steckverbinder und Kabel muss in Abstimmung mit dem Lieferanten der Server Hardware durchgeführt werden. Siehe auch "Schematische Darstellung DSTW-ZE Serverschrank"		
PH Text:	Wird akzeptiert.		

Commented [PA35]: Comment (38) by PAUL Alexander (19 Dez 2025 12:17)
duplicate of REQ-01132.
PAUL Alexander (19 Dez 2025 12:31)
already removed in DOORS.

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EL4-DIL-SSS-REQ-01575 Server Hardware Requirements - Power Supply - (Draft) Cable Colours			
The redundant server power supplies shall be cabled using supply cables of the same colour for each redundancy (for example: red cables are used for the first redundancy and blue cables are used for the second redundancy).			
Comments:			
Category:	GCR	Priority:	Optional
Originator:	Customer	Origin:	=>> [EL4 PH] [20_82]
Target Release:	DIL_OBB_1.0.0	Scope:	System
Allocation:	DIL	Verification:	< will be added in a later document edition >
		See also:	
Issues/Decisions:			
LH Text:	1.7.0-7 Die Verkabelung der redundanten Server-Netzteile soll über farblich einheitliche Versorgungskabel pro Redundanz erfolgen (zum Beispiel: für erste Redundanz werden rote Kabel verwendet, für die zweite Redundanz werden blaue Kabel verwendet).		
PH Text:			

EL4-DIL-SSS-REQ-01133		Server Hardware Requirements - Lockable Doors		(Draft)	
The doors of the server racks shall be lockable with a key.					
Comments:					
Category:	GCR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [20_83]		
Target Release:	DIL_OBB_1.0.0	Scope:	System		
Allocation:	DIL	Verification:	< will be added in a later document edition >		
		See also:			
Issues/Decisions:					
LH Text:		1.7.0-8 Türen der Server-Racks müssen per Schlüssel sperrbar sein.			
PH Text:		Wird akzeptiert.			

EL4-DIL-SSS-REQ-01134 Server Hardware Requirements - Different Locks (Draft)			
The locks on the server racks shall be configured differently. At a minimum, different locks shall be provided for the following applications: → [DIL-CU] AN1Contractor 1 → [DIL-CU] AN2Contractor 2 → Separately for each [DIL] application			
Comments:			
Category:	GCR	Priority:	Mandatory
Originator:	Customer	Origin:	=> [EL4 PH] [20_84]
Target Release:	DIL_OBB_1.0.0	Scope:	System
Allocation:	DIL	Verification:	< will be added in a later document edition >
See also:			
Issues/Decisions:			
LH Text:	1.7.0-9 Die Schlösser der Server-Racks müssen unterschiedlich konfiguriert werden. Mindestens sind für folgende Applikationen unterschiedliche Schlösser vorzusehen: • DSTW-ZE AN1 • DSTW-ZE AN2 • für jede DSTW-Applikation separat		
PH Text:	Wird akzeptiert.		

Commented [PA36]: Proposed Change (39) by PAUL Alexander (19 Dez 2025 12:18)

PAUL Alexander (19 Dez 2025 12:30) already changed in DOORS.

Commented [PA37]: Proposed Change (40) by PAUL Alexander (19 Dez 2025 12:18)

PAUL Alexander (19 Dez 2025 12:30) already changed in DOORS.

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EL4-DIL-SSS-REQ-01135		Network & Transmission Technology - Server to (Draft)	
Network Connection			
The servers shall be connected to the network using: → 1000BASE-T ports for speeds up to 1 GBit/s → SFP+ ports (either DAC or optical) for speeds up to 10 Gbit/s			
Comments:	The following assumptions are made regarding the network requirements at a [DC] location: → The network cabling between the switch and the telecommunications room is installed by the interior designer. → The network cabling between the switch and the server will be installed by the supplier/provider of the server.		
Category:	GCR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [20_86]
Target Release:	DIL_OBB_1.0.0	Scope:	System
Allocation:	DIL	Verification:	< will be added in a later document edition >
		See also:	
Issues/Decisions:			
LH Text:	1.8.0-1 Die Anbindung der Server ans Netzwerk muss mittels: • 1000BASE-T Ports für Geschwindigkeiten bis 1 GBit/s • SFP+ Ports (wahlweise DAC oder optisch) für Geschwindigkeiten bis 10 GBit/s erfolgen. Für die Anforderungen an das Netzwerk an einem Rechenzentrum Standort werden folgende Annahmen zu Grunde gelegt: • Die Netzwerkverkabelung zwischen Switch und TK-Raum wird durch den Raumausstatter installiert. • Die Netzwerkverkabelung zwischen Switch und Server wird durch den Lieferant/Beisteller des Servers installiert.		
PH Text:	Wird akzeptiert.		

EL4-DIL-SSS-REQ-01136		Network & Transmission Technology - BMC Connection		(Draft)	
The baseboard management controller (BMC) shall be connected via a 1000BASE-T port.					
Comments:					
Category:	GCR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [20_87]		
Target Release:	DIL_OBB_1.0.0	Scope:	System		
Allocation:	DIL	Verification:	< will be added in a later document edition >		
		See also:			
Issues/Decisions:					
LH Text:		1.8.0-2 Die Anbindung des Basebord Management Controller (BMC) muss über 1000BASE-T Port erfolgen.			
PH Text:		Wird akzeptiert.			

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EL4-DIL-SSS-REQ-01137 Network & Transmission Technology - [SiDiNet] (Draft) and t[DN2020]			
At least one port shall be provided per server for the [SiDiNet] channel red, [SiDiNet] channel blue and the [DN2020]. The necessary speeds and additional redundancy shall be agreed upon.			
Comments:			
Category:	GCR	Priority:	Mandatory
Originator:	Customer	Origin:	=> [EL4 PH] [20_88]
Target Release:	DIL_OBB_1.0.0	Scope:	System
Allocation:	DIL	Verification:	< will be added in a later document edition >
See also:			
Issues/Decisions:			
LH Text:	1.8.0-3 Pro Server muss zumindest je ein Port für den SiDiNet Kanal Rot, SiDiNet Kanal Blau und das DN2020 vorgesehen werden. Die notwendigen Geschwindigkeiten, sowie eine zusätzliche Redundanz ist abzustimmen.		
PH Text:	Details dieser Vorgabe müssen in den Pflichtenheftkunden besprochen werden.		

EL4-DIL-SSS-REQ-01139 Network & Transmission Technology - (Draft) Separation of Transmission Technology and Data Network			
Separate rooms shall be provided for the security technology for the applications of the core components of the transmission technology and the data network in order to enable separate access control.			
Comments:			
Category:	GCR	Priority:	Mandatory
Originator:	Customer	Origin:	=> [EL4 PH] [20_91]
Target Release:	DIL_OBB_1.0.0	Scope:	System
Allocation:	DIL	Verification:	< will be added in a later document edition >
See also:			
Issues/Decisions:			
LH Text:	1.8.0-6 Für die Applikationen der Core-Komponenten der Übertragungstechnik und des Datennetzes müssen separate Räumlichkeiten zur Sicherungstechnik zur Verfügung gestellt werden, um eine getrennte Zugangskontrolle zu ermöglichen.		
PH Text:	Wird akzeptiert.		

EL4-DIL-SSS-REQ-01140 Network & Transmission Technology - Location (Draft) of Premises			
In order to be able to carry out possible future expansions, the premises for network and transmission technology shall be located in the same fire compartment as the premises for security technology.			
Comments:			
Category:	GCR	Priority:	Mandatory
Originator:	Customer	Origin:	=> [EL4 PH] [20_92]
Target Release:	DIL_OBB_1.0.0	Scope:	System
Allocation:	DIL	Verification:	< will be added in a later document edition >
See also:			
Issues/Decisions:			
LH Text:	1.8.0-7 Um mögliche Erweiterungen in der Zukunft durchführen zu können, müssen Räumlichkeiten der Netzwerk- & Übertragungstechnik im selben Brandabschnitt liegen, wie die Räumlichkeiten der Sicherungstechnik.		
PH Text:	Wird akzeptiert.		

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EL4-DIL-SSS-REQ-01141		Network & Transmission Technology - Redundant Power Supplies		(Draft)
Redundant power supplies for the network and transmission technology components shall be supplied from independently secured power circuits.				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_93]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.8.0-8 Redundante Netzteile der Netzwerk- & Übertragungstechnik-Komponenten müssen aus unabhängig voneinander abgesicherten Stromkreisen versorgt werden.			
PH Text:	Wird akzeptiert.			

EL4-DIL-SSS-REQ-01142		Network & Transmission Technology - Separation of SiDiNet and Data Network		(Draft)
The following network infrastructures shall be physically separated in the [DC]:				
→ [SiDiNet] channel "red"				
→ [SiDiNet] channel "blue"				
→ Data network 2020				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [20_95]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.8.0-10 Folgende Netzwerkinfrastrukturen müssen im Rechenzentrum physikalisch getrennt werden: • SiDiNet Kanal "rot" • SiDiNet Kanal "blau" • Datennetz 2020			
PH Text:	Details dieser Vorgabe müssen in den Pflichtenheftkunden besprochen werden.			

EL4-DIL-SSS-REQ-01576		Network & Transmission Technology - Network Cabling		(Draft)
The network cabling of the server hardware should be carried out using network cables of a uniform colour (for example: red network cables are used for [SiDiNet] channel "red", blue network cables are used for [SiDiNet] channel "blue").				
Comments:				
Category:	GCR	Priority:	Optional	
Originator:	Customer	Origin:	==> [EL4 PH] [20_97]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
See also:				
Issues/Decisions:				
LH Text:	1.8.0-12 Die Netzverkabelung der Server Hardware soll über farblich einheitliche Netzwirkkabel erfolgen (zum Beispiel: für SiDiNet Kanal "rot" werden rote Netzwirkkabel verwendet, für SiDiNet Kanal "blau" werden blaue Netzwirkkabel verwendet).			
PH Text:				

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3.2 NETWORKS

This chapter is derived from the customer specification [PH_PART_D6].
The requirements in this chapter consider the Security Service Network ([SiDiNet]), consisting of

- Track field network,
- Transmission technology ([ÜT]), and
- Network data center

For the concept concerning the track field network and the network data center refer to [Network_Concept_SiDiNet].

3.2.1 SEPARATION OF INTERFACES

EL4-DIL-SSS-REQ-00543		Networks - Interfaces Redundant No VLAN Separation per EULYNX		(Draft)
All interfaces ([SCI], [SMI], [SDI] and [SSI]) shall be realized with redundancy (two physical [PoS] signaling interfaces). The [SCI], [SMI], [SDI] and [SSI] interfaces are not separated based on [VLAN] (see [EULYNX Specification] - (Eu.Pos.1143)).				
Comments:		- Both [OS] and the [DIL-CU] servers are equipped with several network interfaces. - The PoS signaling interface is the switch port on the first network device that is part of the [SiDiNet] to which the cable of the EULYNX system or an adjacent system is connected (see [Eu.Pos.980]).		
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [22_18] ==> [EL4 PH] [22_19] ==> [EL4 PH] [22_20]	
Target Release:	DIL_OBB_1.0.0 DIL_OBB_1.0.x DFE_1.0.0 DCA_OBB_3.1.0 AzLM_OBB_v7.10	Scope:	System	
Allocation:	DIL, IPS, DFE	Verification:	< will be added in a later document edition >	
Issues/Decisions:		See also:		
LH Text:	1.1.0-1 Alle Schnittstellen (SCI, SMI, SDI und SSI) müssen redundant ausgeführt werden (zwei physikalische PoS-Signalling-Schnittstellen). 1.1.0-2 Die PoS-Signalling-Schnittstelle ist der Switchport am ersten Netzwerkgerät, der Teil des SiDiNet ist, an dem das Kabel des EULYNX-Systems oder eines benachbarten Systems angeschlossen wird. Siehe Eu.Pos.980 [145]. 1.1.0-3 Die SCI-, SMI-, SDI-, SSI-Schnittstellen werden untereinander nicht VLAN-basiert getrennt (No separation). Siehe Eu.Pos.1143 [145].			
PH Text:	Die Object Controller als auch die Server DSTW-ZE besitzen mehrere Netzwerkschnittstellen, womit die Anforderung von physikalisch getrennten Schnittstellen umgesetzt wird.			

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EL4-DIL-SSS-REQ-00544 Networks - Interface Traffic Prioritized via DSCP (Draft) per EULYNX			
The prioritization [QoS] of the network traffic of the [SCI], [SMI], [SDI] and [SSI] interfaces shall be done using [DSCP] markers according to [EULYNX Specification] - (Eu.Pos.1140).			
Comments:			
Category:	FUR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [22_21]
Target Release:	DIL_OBB_1.0.0	Scope:	System
	DIL_OBB_1.0.x		
	DCA_OBB_3.1.0		
	AzLM_OBB_v7.10		
Allocation:	DIL, IPS, DFE	Verification:	< will be added in a later document edition >
		See also:	
Issues/Decisions:			
LH Text:	1.1.0-4 Die Priorisierung (QoS) des Netzwerktraffics der SCI-, SMI-, SDI-, SSI-Schnittstellen muss mittels DSCP-Marker erfolgen. Siehe Eu.Pos.1140 [145].		
PH Text:	Eine Implementierung der Priorisierung des Traffics anhand von DSCP-Markern wird primär von den Netzwerkkomponenten behandelt und umgesetzt. Soweit diese im Lieferumfang des Auftragnehmers stehen, werden diese auch darauf Rücksicht nehmen.		

EL4-DIL-SSS-REQ-00545 Networks - Sender Marks Interface Traffic with (Draft) DSCP			
The traffic on the [SCI], [SMI], [SDI] and [SSI] interfaces shall be provided with a [DSCP] traffic marker by the sender.			
Comments:			
Category:	FUR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [22_22]
Target Release:	DIL_OBB_1.0.0	Scope:	System
	DIL_OBB_1.0.x		
	DCA_OBB_3.1.0		
	AzLM_OBB_v7.10		
Allocation:	DIL, IPS, DFE	Verification:	< will be added in a later document edition >
		See also:	
Issues/Decisions:			
LH Text:	1.1.0-5 Der Traffic auf der SCI-, SMI-, SDI- und SSI-Schnittstelle muss vom Absender mit einem DSCP Traffic Marker versehen werden.		
PH Text:	Das Setzen der DSCP Marker erfolgt im VSTW durch den Object Controller. Eine Vorgabe der zu verwendeten Markerwerte ist vom Auftraggeber zur Verfügung zu stellen.		

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EL4-DIL-SSS-REQ-00546 Networks - Separate Networks shall Use Distinct IP Segments (Draft)			
The physically separate networks <u>with their to which the</u> physically separate [PoS] signaling interfaces are connected shall use different IP segments.			
Comments:			
Category:	FUR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [22_23]
Target Release:	DIL_OBB_1.0.0	Scope:	System
	DIL_OBB_1.0.x		
	DCA_OBB_3.1.0		
	AzLM_OBB_v7.10		
Allocation:	DIL, IPS, DFE	Verification:	< will be added in a later document edition >
		See also:	
Issues/Decisions:			
LH Text:	1.1.0-6 Die physisch getrennten Netze, an denen die physisch getrennten PoS-Signalling-Schnittstellen angeschlossen sind, müssen unterschiedliche IP-Segmente verwenden.		
PH Text:	Eine Konfiguration der IP Adressen ist für VSTWs und Object Controller unterstützt und vorgesehen. Die zu verwendeten IP Adressen sind vom Auftraggeber zu Verwalten und bereitzustellen.		

EL4-DIL-SSS-REQ-00547 Networks - NTP Traffic Marked per EULYNX/EU-Rail SP Interface Spec (Draft)			
[The [NTP] traffic to/from the [ÖBB] time server shall be marked with the traffic marker for the interface specified by EULYNX or EU-Rail SP documents.]			
Comments:			
Category:	FUR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [22_25]
Target Release:	DIL_OBB_1.0.0	Scope:	System
	DIL_OBB_1.0.x		
	DCA_OBB_3.1.0		
	AzLM_OBB_v7.10		
Allocation:	DIL, IPS, DFE	Verification:	< will be added in a later document edition >
		See also:	
Issues/Decisions:			
LH Text:	1.1.0-8 Der NTP-Traffic zum/vom ÖBB-Zeitserver muss mit dem Traffic Marker für die durch EULYNX bzw. EU-Rail SP Dokumente spezifizierte Schnittstelle gekennzeichnet werden.		
PH Text:	Der Einsatz von Trafficmarkern für den Datentrafic zum/vom NTP-Server auf der SSI-Schnittstelle wird unterstützt. Die Information welcher Marker genau zu verwenden ist, ist vom Auftraggeber bereitzustellen.		

Commented [FM38]: Comment (13) by FRANZL Michael (17 Dez 2025 15:05)
SP-SEC requires NTS. NTP only during commissioning.

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EL4-DIL-SSS-REQ-00548		Networks - Customer-Defined DSCP Markers		(Draft)
Must Be Configurable				
The respective [DSCP] traffic markers (set by the customer) shall be configurable.				
Comments:				
• The [DSCP] traffic markers are defined during the realization. • The [QoS] priorities in the [SiDiNet] are set according to the [PoS] signalling.				
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	=> [EL4 PH] [22_26]	
			=> [EL4 PH] [22_27]	
			=> [EL4 PH] [22_29]	
Target Release:	DIL_OBB_1.0.0 DIL_OBB_1.0.x DCA_OBB_3.1.0 AzLM_OBB_v7.10	Scope:	System	
Allocation:	DIL, VIL, IPS, DFE	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.1.0-9 Die jeweiligen DSCP Traffic Marker werden vom AG festgelegt und müssen konfigurierbar sein. 1.1.0-10 Die DSCP Traffic Marker werden im Zuge der Umsetzung festgelegt. 1.1.0-12 QoS-Prioritäten werden im SiDiNet entsprechend PoS-Signalling umgesetzt werden.			
PH Text:	Eine Konfiguration von DSCP Markern wird von VSTW und Object Controller unterstützt. Der Auftragnehmer geht davon aus, dass die Marker nicht für jedes VSTW unterschiedlich sind, womit eine generische Konfigurationsvorgabe für die jeweiligen Schnittstellen erfolgen kann. Sind dennoch unterschiedliche Marker notwendig, so sind diese mit ausreichender Vorlaufzeit dem Auftragnehmer bekannt zu geben.			

3.2.2 PROTOCOL STACK

EL4-DIL-SSS-REQ-00549		Networks - IPv4 Required at OSI Layer 3 per EULYNX		(Draft)
The network layer ([OSI] layer 3) shall use [IPv4] (see [EULYNX Specification] - (Eu.Pos.1107)).				
Comments:				
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [22_31]	
Target Release:	DIL_OBB_1.0.0 DIL_OBB_1.0.x DCA_OBB_3.1.0 AzLM_OBB_v7.10	Scope:	System	
Allocation:	DIL, VIL, IPS, DFE	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.2.0-1 Am Network Layer (OSI Layer 3) muss IPv4 verwendet werden. Siehe Eu.Pos.1107 [145].			
PH Text:	Soweit IP als Protokoll zum Einsatz kommt, wird v4 verwendet. Ein Einsatz von IP ist z. B. nicht für 802.1x vorgesehen.			

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EL4-DIL-SSS-REQ-00550		Networks - No Port Channels or Link Aggregation Protocols		(In negotiation)
No port channels or link aggregation protocols shall be used between the two [PoS] signaling interfaces (see [EULYNX Specification] - (Eu.Doc.15)).				
Comments: == OPEN POINT ==				
The [DIL-CU] servers and the [OC]s do not use port channels or link aggregation protocols for the [PoS] signaling interfaces. However, from an architectural point of view, these protocols are necessary for the other interfaces (e.g. data exchange between [ADM], [VIL] instances)				
Reference Eu.Sec.363 in Eu.Doc.15 had been removed with EULYNX BL4R2. Unclear if customer still require this to be covered.				
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [22_32]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
	DIL_OBB_1.0.x			
	DCA_OBB_3.1.0			
	AzLM_OBB_v7.10			
Allocation:	DIL, VIL, IPS, DFE	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.2.0-2 Zwischen den beiden PoS-Signalling-Schnittstellen dürfen keine Port-Channel oder Link Aggregation Protokolle zum Einsatz kommen. Siehe Eu.Sec.363 [105].			
PH Text:	Die Server der DSTW-ZE als auch Object Controller verwenden für die PoS-Signalling-Schnittstellen keine Port-Channel oder Link Aggregation Protokolle. Jedoch sind aus architektonischer Sicht für die anderen Schnittstellen (z. B. Datenaustausch zwischen ADM, VSTW-Redundanzinstanzen, etc.) teils diese Protokolle notwendig.			

3.2.3 NETWORK GENERAL

The network components as part of the [PoS]-Signalling ([Eu.Doc.100]) do not deliver any cryptographic functionality concerning the transmission.

3.2.4 AVAILABILITY

[SiDiNet] has a defined availability of ~~0.9999799~~ 99.97 %.

3.2.5 TRACK NETWORK

[SiDiNet] corresponds to a network of category 3 according to [EN 50159].

The track network consists of two redundantly constructed rings (channel RED and channel BLUE).

The start and end switches and a spare switch in a track network ring are used internally by the network.

Commented [MF39]: Proposed Change (1) by MOSER Florian (17 Dez 2025 9:43) to be consistent with percentage notation as in other SSS documentation

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EL4-DIL-SSS-REQ-00944		Networks - Switches with DIN Rail Mounting		(Draft)
For all [FES] sizes, switches with [DIN rail] mounting shall be provided.				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [22_50]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
	DIL_OBB_1.0.x			
Allocation:	DIL, DFE	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.5.0-11 Bei allen FES-Größen müssen Switches mit Hutschienenmontage befestigt werden können.			
PH Text:	Für die FES der Größe "S" und "M" sind Hutschienen installiert. Für "L" bis "XXL" können optional zur 19"-Ausführung Hutschienen für die Switchmontage installiert werden.			

EL4-DIL-SSS-REQ-00945		Networks - Switches in 19" Design		(Draft)
For [FES-L] up to [FES-XXL], it shall be possible to attach additional switches in 19" design.				
Comments:				
Category:	MCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [22_51]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
	DIL_OBB_2.0.0			
Allocation:	DIL, DFE	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.5.0-12 Bei den FES-Größen L - XXL müssen zusätzlich Switches in 19"-Ausführung befestigt werden können.			
PH Text:	Für FES ab Größe L sind 19" Befestigungsrahmen installiert.			

3.2.6 DN2020 NETWORK

< intentionally left blank >

3.2.7 NETWORK DATA CENTER

The network connections for channel Red+A, Blue+B and [DN2020] are provided by a physically separate switch infrastructure.

In the [DC], the [DIL] network segments and other networks (e.g. [X25oRaSTA] network), use the same physical hardware. They are separated from each other via [VLAN]s.

Data traffic in the network can be diverted using a mirror port or optical [TAP].

The typical bit error rate [BER] of [DC] connections is 1E-13.

The support of other protocols (e.g. [MPLS]) in the [DC] network is not required.

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EL4-DIL-SSS-REQ-00551		Network [DC] - General Requirements		(Draft)
The network between the [DC] locations shall fulfill the following requirements from the contractor's perspective:				
→ Required/guaranteed data rate: 50 Gbit/s (DSTW-ZE – DSTW-ZE connection in total)				
→ Maximum delay between sender and recipient: 15 ms				
→ Permissible jitter between redundant lines: 4 ms				
→ Permissible jitter between sender and receiver: 4.5 ms (30%)				
→ Permissible bit error rate per redundant network: BER = 1E-15.				
Comments:				
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [22_67]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL, IPS	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.7.0-3 Die notwendige Bandbreite zwischen den Rechenzentrum-Standorten muss bekannt gegeben werden, die Möglichkeit der Umsetzung muss mit dem AG abgestimmt werden und die Bandbreitenberechnung muss bekanntgegeben werden.			
PH Text:	Der AN erwartet die folgenden Netzwerkanforderungen zwischen den Rechenzentrum-Standorten: Geforderte/Garantierte Daten Rate: 50 Gbit/s (DSTW-ZE – DSTW-ZE Verbindung gesamt) - Maximales Delay zwischen Sender – Empfänger: 15 ms - Erlaubter Jitter zwischen den Redundanten Leitungen: 4 ms Erlaubter Jitter zwischen Sender – Empfänger: 4,5 ms (30%) - Erlaubte Bit-Fehlerrate pro redundantes Netzwerk: BER = 1E-13			

EL4-DIL-SSS-REQ-00552		Network [DC] - Routable Network Connections with Track Field Network		(Draft)
All network-based connections between the track field network and the [DC]network shall be routable ([OSI] Layer 3).				
Comments:		This addresses the network-based connections to the [OC]s.		
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [22_69]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL, IPS	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.7.0-5 Alle netzwerkbasieren Verbindungen zwischen Gleisfeldnetzwerk und Netzwerk Rechenzentrum müssen routingfähig sein (OSI Layer 3).			
PH Text:	Sämtliche Verbindungen zu den Object Controllern sind routingfähig.			

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EL4-DIL-SSS-REQ-00553		Network [DC] - Routable Network Connections (Draft) between DC Locations	
All network-based connections between the [DC] locations shall be routable ([OSI] Layer 3).			
Comments:			
Category:	FUR	Priority:	Mandatory
Originator:	Customer	Origin:	=> [EL4 PH] [22_71]
Target Release:	DIL_OBB_1.0.0	Scope:	System
Allocation:	DIL, IPS	Verification:	< will be added in a later document edition >
		See also:	
Issues/Decisions:			
LH Text:	1.7.0-7 Alle netzwerkbasierten Verbindungen zwischen den Rechenzentrums-Standorten müssen routingfähig sein (OSI Layer 3).		
PH Text:	Alle netzwerkbasierten Verbindungen zwischen den Rechenzentrums-Standorten sind routingfähig.		

EL4-DIL-SSS-REQ-00554		Network [DC] - Storage Network - Coordination		(Draft)
If necessary, the supplier shall coordinate the requirements for the storage networks with the customer (bandwidth, protocols, interfaces, hardware).				
Comments:		The storage networks are not included in the [LH] D6.		
Category:	MCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [22_72]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
	DIL_OBB_1.0.x			
Allocation:	DIL, DFE	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.7.0-8 Storagenetze müssen bei Bedarf abgestimmt werden, diese sind nicht im Lastenheft Netze erfasst. (Bandbreite, Protokolle, Schnittstellen, Hardware)			
PH Text:	Der AN wird bei Bedarf die Anforderungen an die Storagenetze mit dem AG abstimmen.			

EL4-DIL-SSS-REQ-00555		Network [DC] - Redirection of Network Traffic - Coordination		(Draft)	
If network traffic needs to be redirected, the supplier shall coordinate this with the customer where necessary.					
Comments:					
Category:	MCR	Priority:	Mandatory		
Originator:	Customer	Origin:	=> [EL4 PH] [22_80]		
Target Release:	DIL_OBB_1.0.0	Scope:	System		
Allocation:	DIL, IPS	Verification:	< will be added in a later document edition >		
		See also:			
Issues/Decisions:					
LH Text:	1.7.0-16 Eine Ausleitung des Datenverkehrs im Netzwerk muss bei Bedarf mit dem AG abgestimmt werden.				
PH Text:	Sollte Bedarf an der Ausleitung des Datenverkehrs im Netzwerk bestehen, wird der AN dies anlassbezogen mit dem AG abstimmen.				

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EL4-DIL-SSS-REQ-01009		Network [DC] - Path Redundancy		(Draft)	
Only [LAG] 802.3ad shall be used for path redundancy in the [DC] network.					
Comments:					
Category: GCR		Priority: Mandatory			
Originator: Customer		Origin: ==> [EL4 PH] [22_141]			
Target Release: DIL_OBB_1.0.0		Scope: System			
Allocation: DIL, IPS		Verification: < will be added in a later document edition >			
		See also:			
Issues/Decisions:					
LH Text:		1.7.0-18 Im Rechenzentrumsnetzwerk darf ausschließlich LAG 802.3ad für eine Wegeredundanz verwendet werden.			
PH Text:		Das Netzwerkkonzept muss während der Pflichtenheftphase im Detail abgestimmt werden.			

EL4-DIL-SSS-REQ-01010		Network [DC] - Prioritisation		(Draft)	
Only 802.1D and DiffServ (RFC 2474, 2475, 2597, 3168, 3246) shall be used for prioritisation with [QoS] in the [DC] network.					
Comments:					
Category:	GCR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [22_143]		
Target Release:	DIL_OBB_1.0.0	Scope:	System		
Allocation:	DIL, IPS	Verification:	< will be added in a later document edition >		
		See also:			
Issues/Decisions:					
LH Text:		1.7.0-19 Im Rechenzentrumsnetzwerk dürfen ausschließlich 802.1D und DiffServ (RFC 2474, 2475, 2597, 3168, 3246) für Priorisierung mit QoS verwendet werden.			
PH Text:		Das Netzwerkkonzept muss während der Pflichtenheftphase im Detail abgestimmt werden.			

3.2.8 TRANSMISSION TECHNOLOGY

The latency between a port on the track field network switch and a port in the [DC] has usually a maximum of 50 ms.

3.2.9 TIME SYNCHRONIZATION

EL4-DIL-SSS-REQ-00557		Time Synchronisation - NTP Time Server - Usage		(Draft)
All [DIL] devices shall use the [ÖBB] [NTP] time server with the [IP] address 10.0.0.11 on the [SSI] interface.				
Comments:		Up to two [NTP] servers can be configured.		
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [22_96]	
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL, VIL, IPS, DFE	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.10.0-2 Alle DSTW-Geräte müssen den oder die ÖBB-NTP-Zeitserver auf der in EULYNX bzw. System Pillar spezifizierten Schnittstelle verwenden.			
PH Text:	Es können bis zu zwei NTP-Server konfiguriert werden.			

Commented [FM40]: Comment (14) by FRANZL Michael (17 Dez 2025 15:07)
Needs updating according to the cited LH text.

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EL4-DIL-SSS-REQ-00560		Time Synchronisation - NTP Time Server - Configuration		(Draft)
The [IP] address of the external [NTP] server shall be configurable.				
Comments:				
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:		
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL, VIL, IPS, DFE	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	1.10.0-2 Alle DSTW-Geräte müssen den oder die ÖBB-NTP-Zeitserver auf der in EULYNX bzw. System Pillar spezifizierten Schnittstelle verwenden.			
PH Text:				

EL4-DIL-SSS-REQ-00561		Time Synchronisation - Internal Time Synchronisation		(Draft)
In case no external time service is available, all time-related system functions shall remain synchronised relative to each other. The transition from external to internal synchronisation shall not lead to a time jump greater than 100 milliseconds.				
Comments:		Note that time stamps of messages generated outside the [VIL] (e.g. host server diagnosis data) shall also be synchronised to the common time base.		
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:		
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL, VIL, IPS	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:				
PH Text:				

EL4-DIL-SSS-REQ-00562		Time Synchronisation - Interface		(Draft)
The system shall connect to the external [NTP] service using				
• [NTP] version 4 • Transport protocol [UDP] via port 123				
Comments:				
Category:	FUR	Priority:	Mandatory	
Originator:	Customer	Origin:		
Target Release:	DIL_OBB_1.0.0	Scope:	System	
Allocation:	DIL	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:				
PH Text:				

Commented [RC41]: Comment (27) by RUGGENTHALER Christoph (18 Dez 2025 16:24)
IPS

3.2.10 IP NETWORK PLANNING GUIDELINES

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3.2.11 MAINTENANCE WORKS

EL4-DIL-SSS-REQ-00558		Maintenance Works - Certificate Renewal Process		(Draft)	
The contractor shall implement the certificate renewal process in such a way that there is no interruption of the [RaSTA] connection.					
Comments:		A connection interruption can occur, for example, if the [TLS] connection has to be reestablished to apply the new certificate.			
Category:	FUR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [22_108]		
Target Release:	DIL_OBB_1.0.0		==> [EL4 PH] [22_109]		
Allocation:	DIL, IPS	Scope:	System		
		Verification:	< will be added in a later document edition >		
		See also:			
Issues/Decisions:					
LH Text:		1.12.0-4 Der AN muss den Zertifikatserneuerungsprozess so implementieren, dass es dabei zu keiner Unterbrechung der RaSTA-Verbindung kommt.			
		1.12.0-5 Eine Verbindungsunterbrechung kann z.B. auftreten, wenn die TLS-Verbindung neu aufgebaut werden muss, um das neue Zertifikat anzuwenden.			
PH Text:		Der Wechsel von Zertifikaten wird unterbrechungsfrei von statten gehen.			

3.2.12 SEPARATION OF NETWORK AND SECURITY SYSTEM

EL4-DIL-SSS-REQ-00559		Separation of Network and Security System - SiDiNet		(Draft)	
As the [SiDiNet] is not part of the security system it shall not be part of the security case.					
Comments:		The [SiDiNet] is considered a gray channel.			
Category:	FUR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [22_114]		
Target Release:	DIL_OBB_1.0.0	Scope:	System		
Allocation:	DIL, VIL, IPS, DFE	Verification:	< will be added in a later document edition >		
		See also:			
Issues/Decisions:					
LH Text:	1.13.0-1 Das SiDiNet ist kein Teil der Sicherungsanlage und darf daher auch nicht Teil des Sicherheitsnachweises sein.				
PH Text:	Das SiDi-Net wird als grauer Kanal betrachtet, womit das SiDi-Net nicht Teil des Sicherheitsnachweises sein muss.				

Commented [BR42]: Comment (10) by BUEGER Roman (17 Dez 2025 14:51)
Is it really "security system" resp. "Security case"?
Usually "Sicherheitsnachweis" is a safety case.
RADDA Klaus (17 Dez 2025 15:03)
Security case would fit as such a document is planned (e.g. its defined in TS 50701 whats in it). of course it starts earlier (e.g. Security concept).
Instead security system I would change with "system boundary", then it is more generic.

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4. HISTORY

4.1 EDITION 01

4.1.1 PROPOSAL 01

This is the first version of the document.

This version covers the **Data Center** requirements according to [LH_PART_D] D4 "Rechenzentrum" and the **Network** requirements according to [LH_PART_D] D6 "Netze".

Please consider the following review guidelines for the review of the document:

- The **allocation** of the requirements to subsystems and components will be reviewed separately. Feel free to make your comments already now.
- The **verification measures** of the requirements will also be reviewed separately. Feel also free to make your comments already now.
- **References and abbreviated terms** are marked with square brackets. They will be later replaced by hyperlinks (after the transfer to Polarion™).

The review **shall not result in a collection of hundreds of medium and grave comments**. If serious objections concerning the current structure and the content of the document exist, please let us clarify this better in an online discussion.

4.1.2 RELEASED

Version after Review.

A complete list of the differences can be found in the attached document "1258_Diff_Ed01Pr01_Ed01RL".

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APPENDIX A DECISIONS

This chapter shall contain important decisions made in connection with the specified requirements.

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